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Kim et al.

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(54) **VEHICLE HEAT PUMP SYSTEM**

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B60H 1/03 (2006.01)
B60H 1/00 (2006.01)

(52) **U.S. Cl.**
CPC **B60H 1/00342** (2013.01); **B60H 1/00278** (2013.01); **B60H 1/03** (2013.01); **B60H 2001/00307** (2013.01)

(58) **Field of Classification Search**

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Primary Examiner — Tavia Sullens

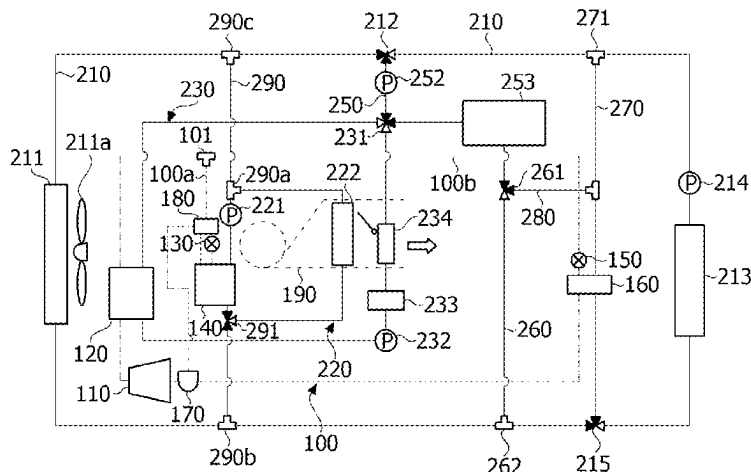
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(57) **ABSTRACT**

An embodiment of the present invention provides a vehicle heat pump system including a compressor configured to compress and circulate a refrigerant, a condenser configured to condense the compressed refrigerant, a first expansion valve configured to expand the condensed refrigerant, an evaporator configured to evaporate the refrigerant expanded by the first expansion valve by allowing the refrigerant to exchange heat with a coolant, a cabin cooler configured to cool a vehicle interior by allowing the coolant having passed through the evaporator to exchange heat with air, and a radiator configured to allow the coolant for cooling an electrical component to exchange heat with outside air, in

(Continued)



which the coolant having passed through the radiator flows into the evaporator in a heating mode.

10 Claims, 25 Drawing Sheets

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(58) **Field of Classification Search**

CPC B60H 1/00921; B60H 2001/00935; B60H 1/143; B60H 1/00899; F25B 2400/13
See application file for complete search history.

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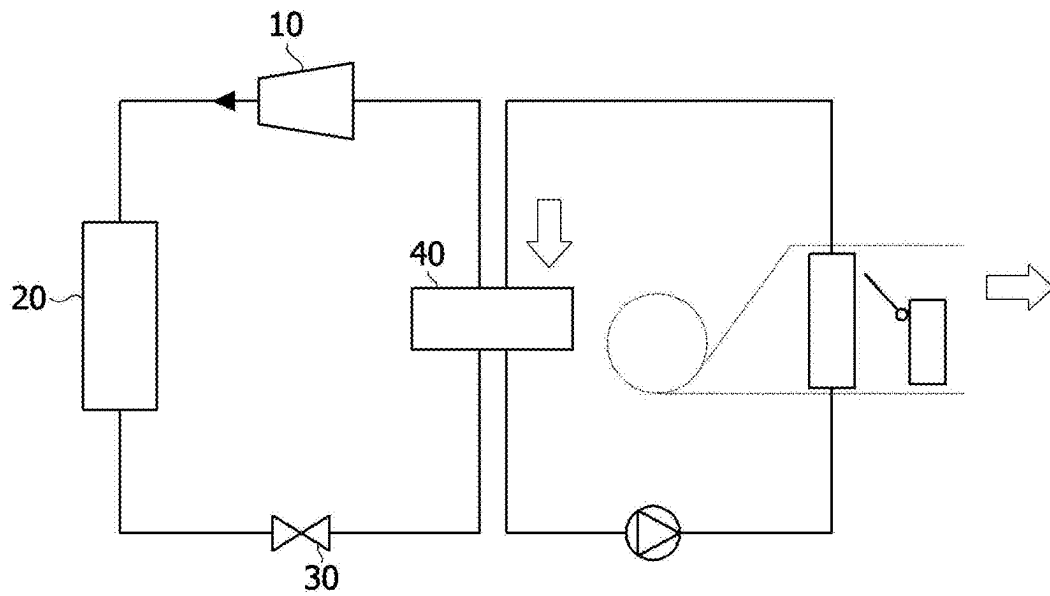
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FIG. 1



PRIOR ART

FIG. 2

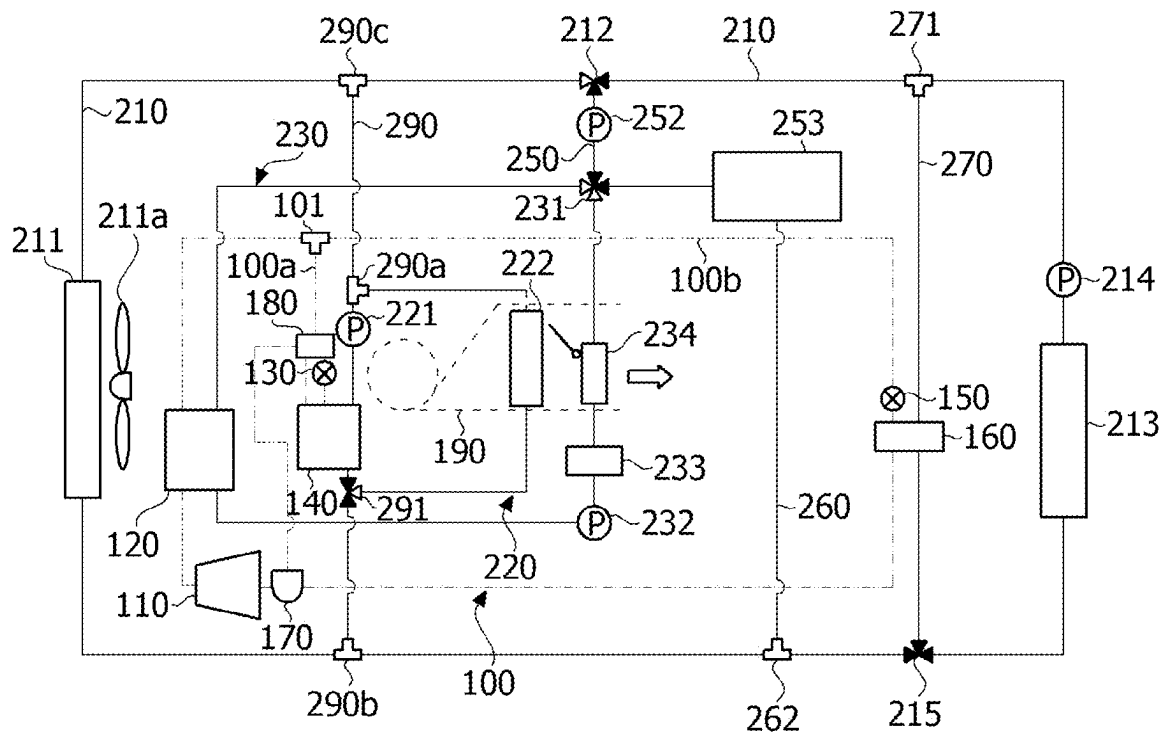


FIG. 3

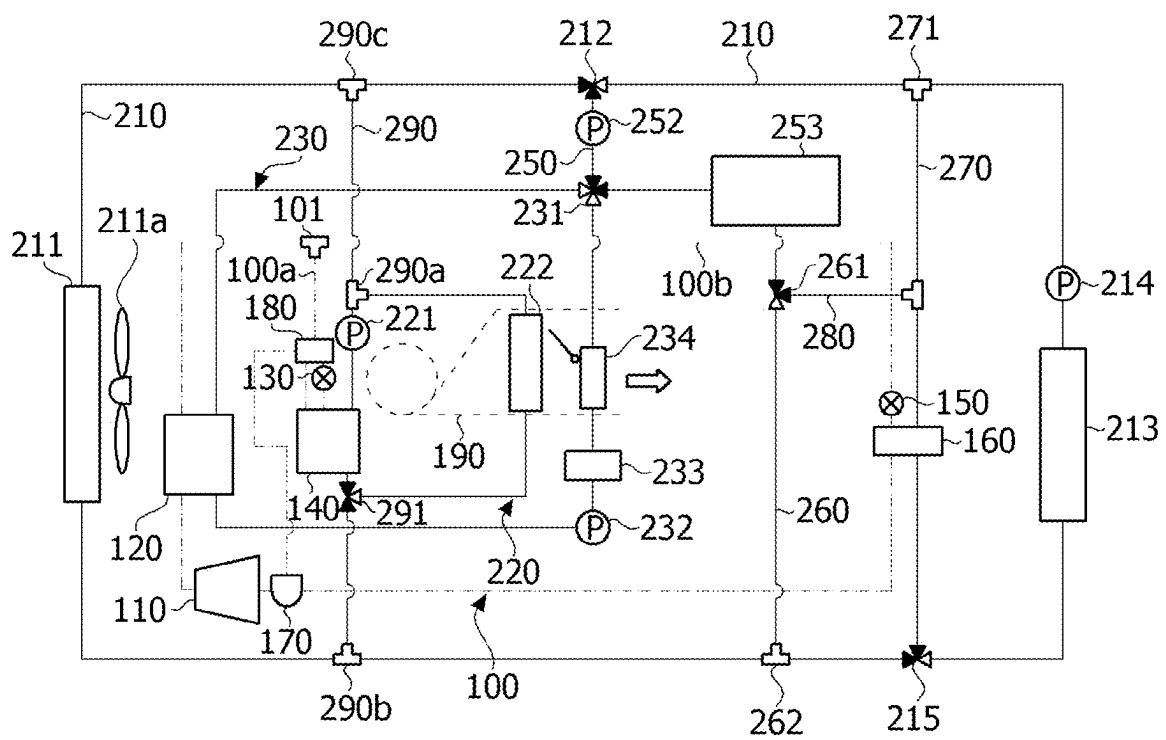
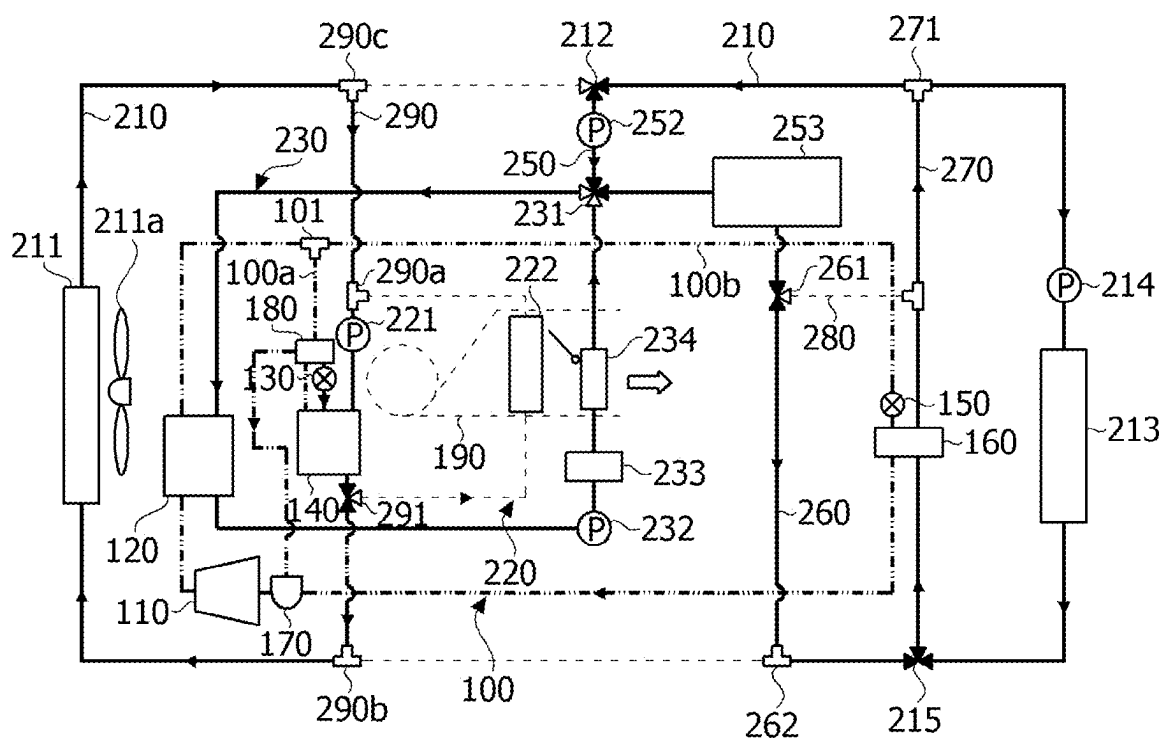


FIG. 4



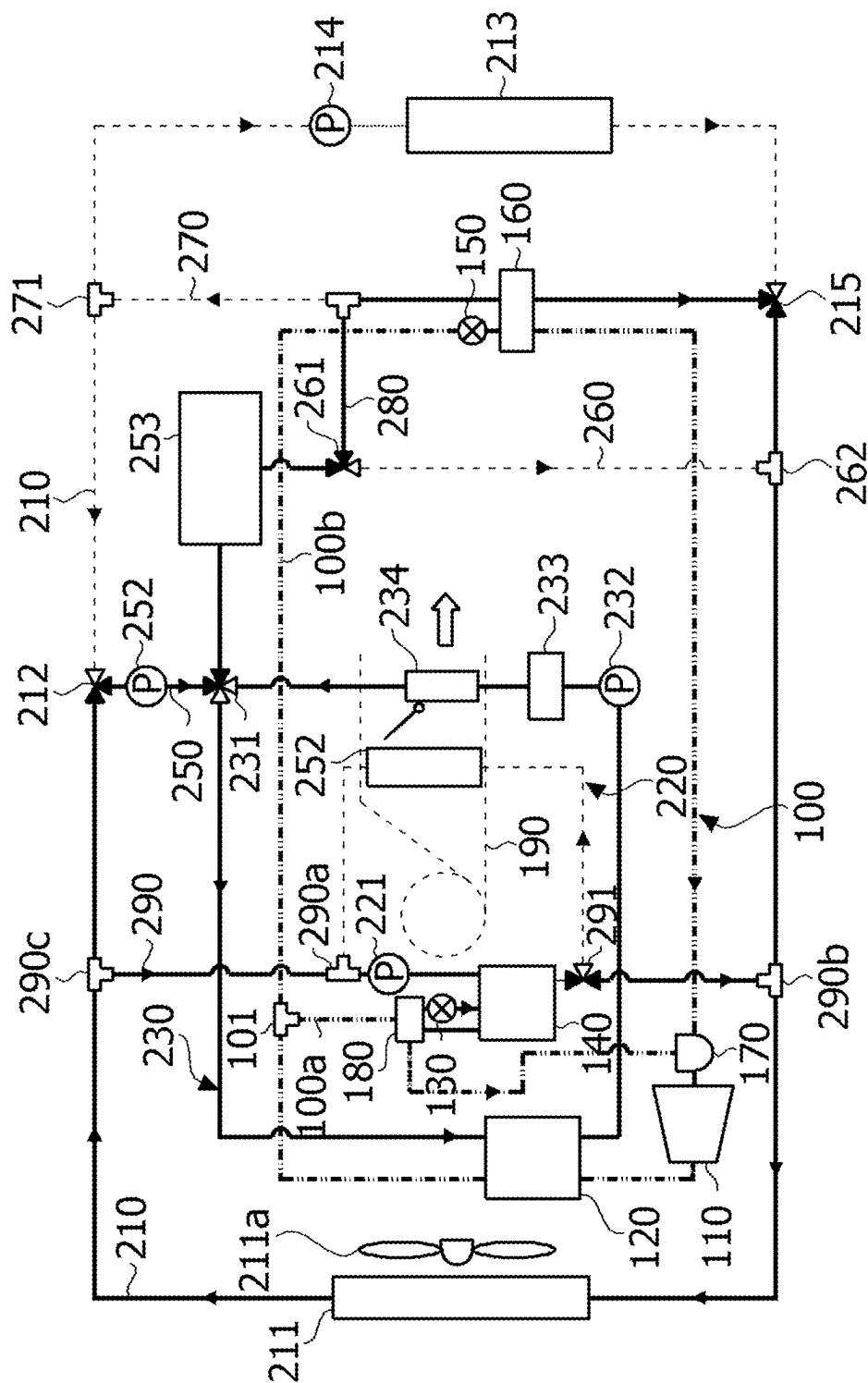


FIG. 5

FIG. 6

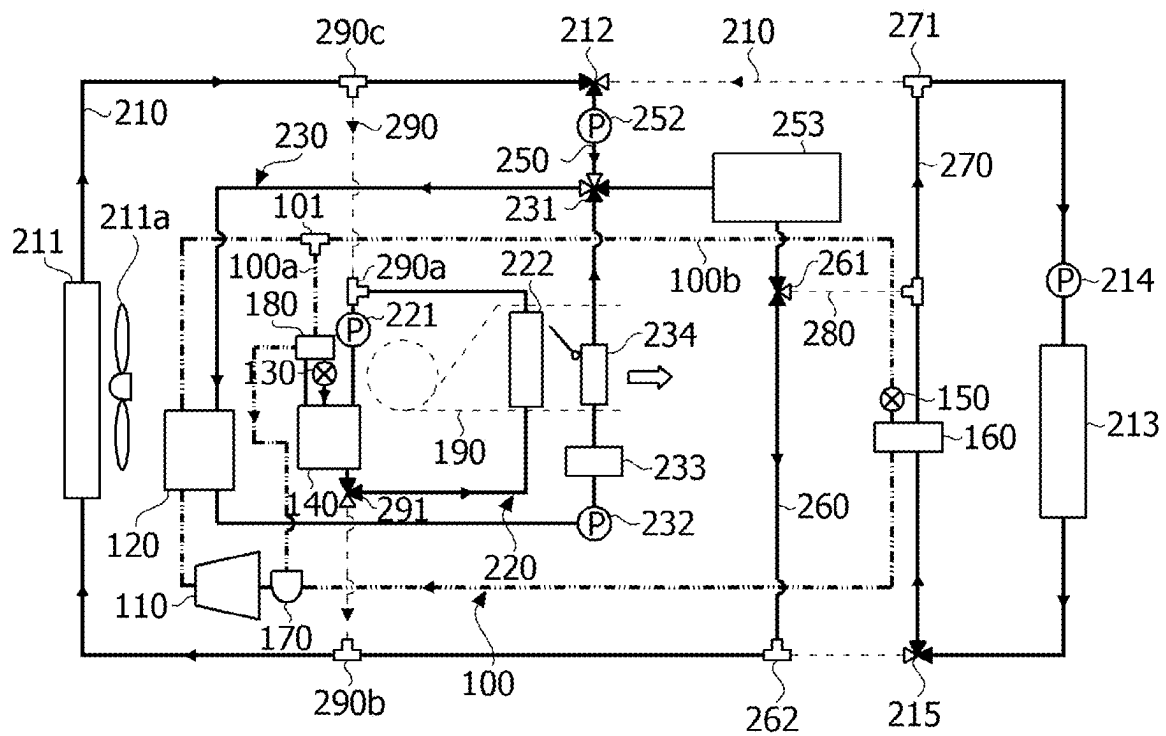


FIG. 7

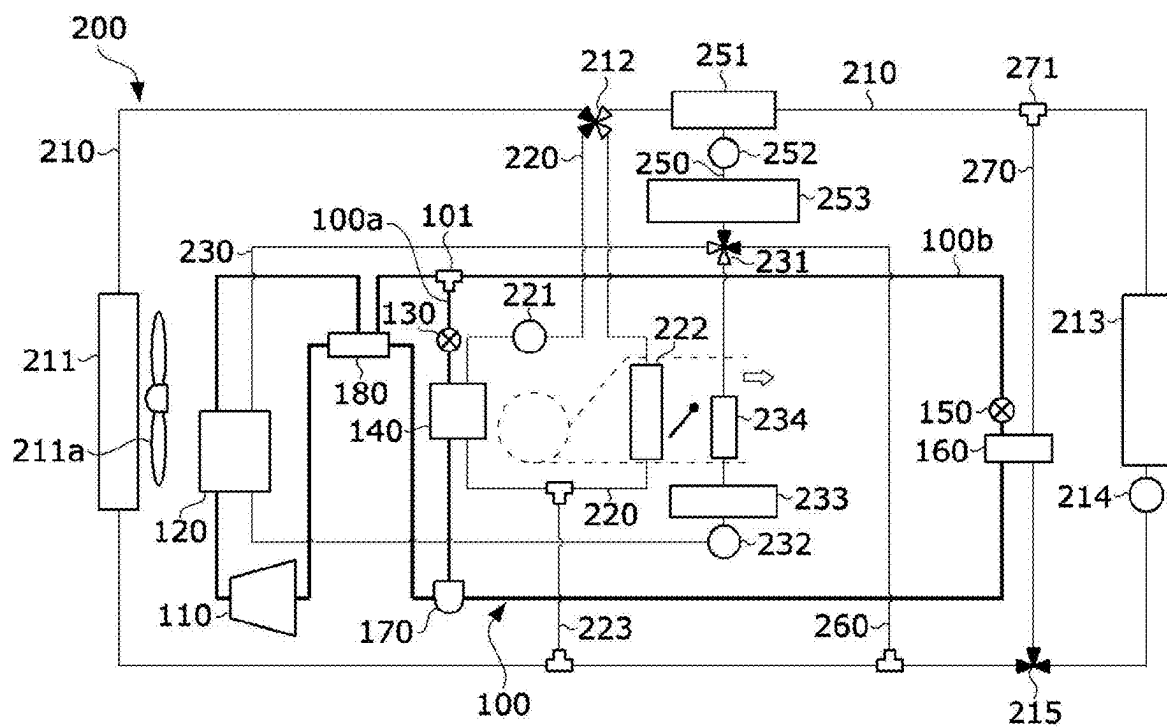


FIG. 8

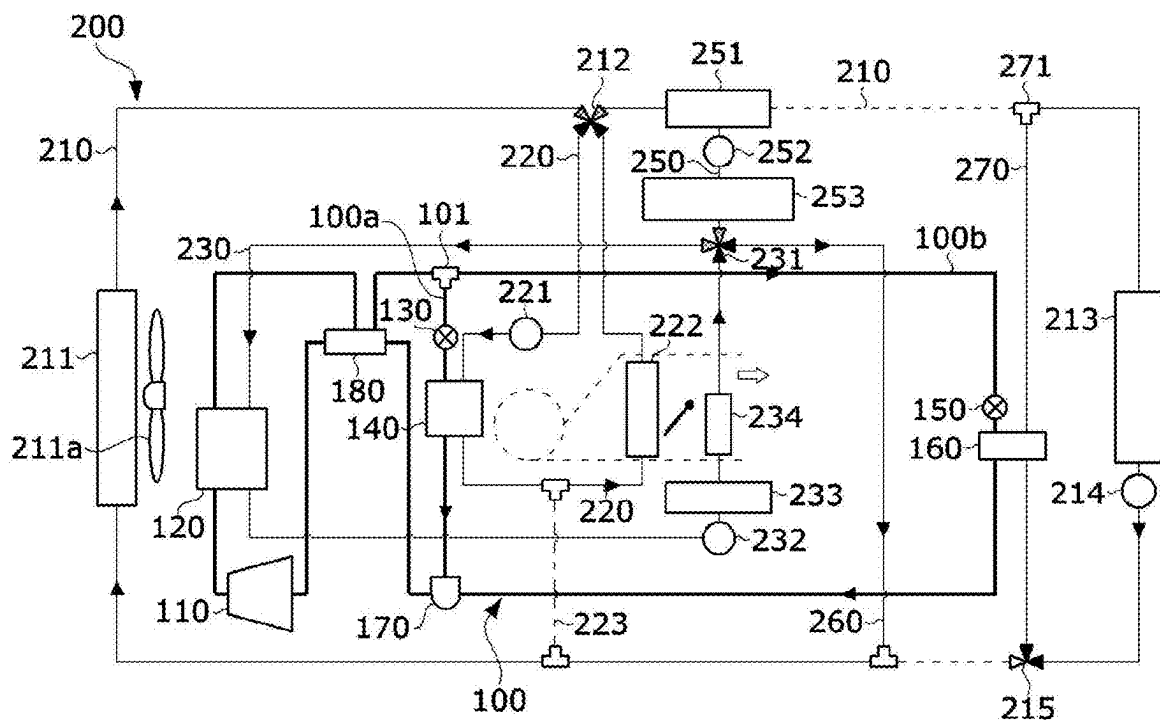


FIG. 9

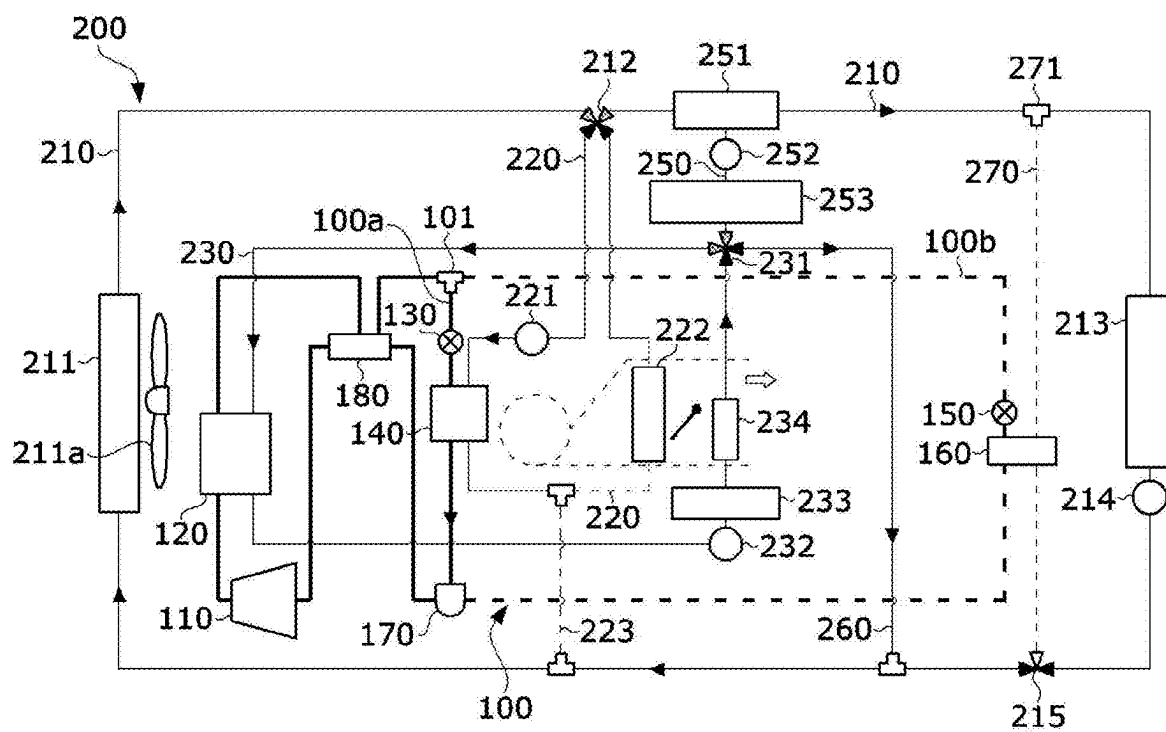


FIG. 10

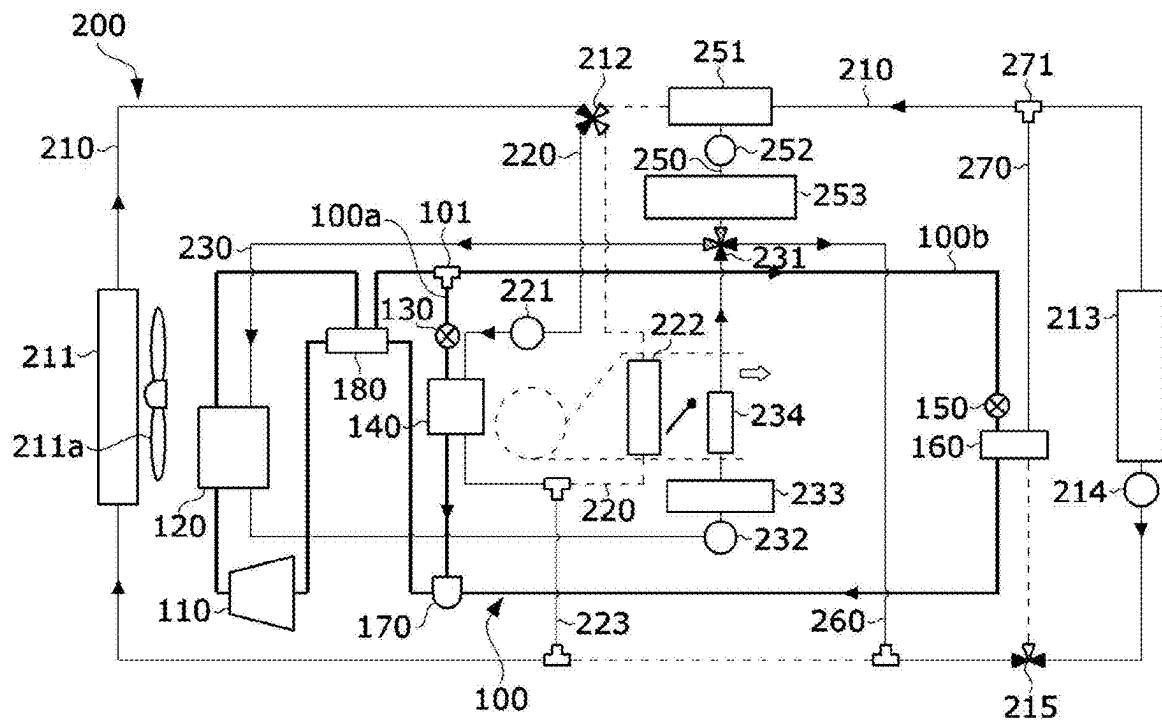


FIG. 11

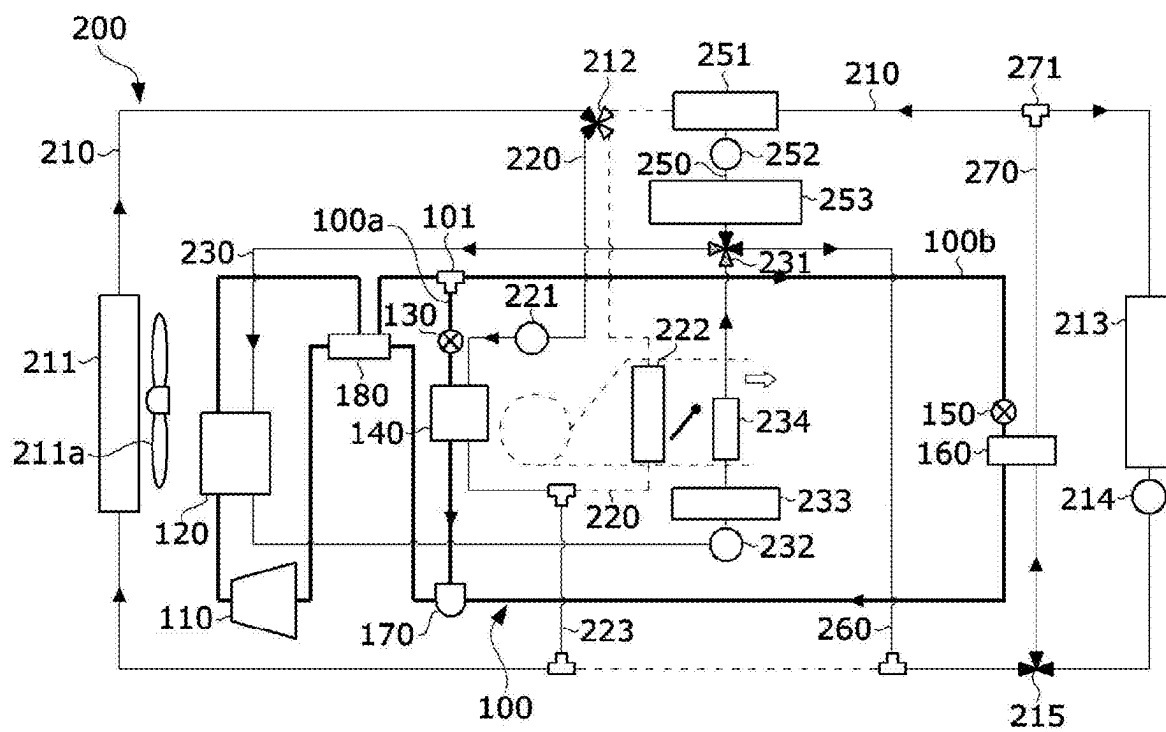


FIG. 12

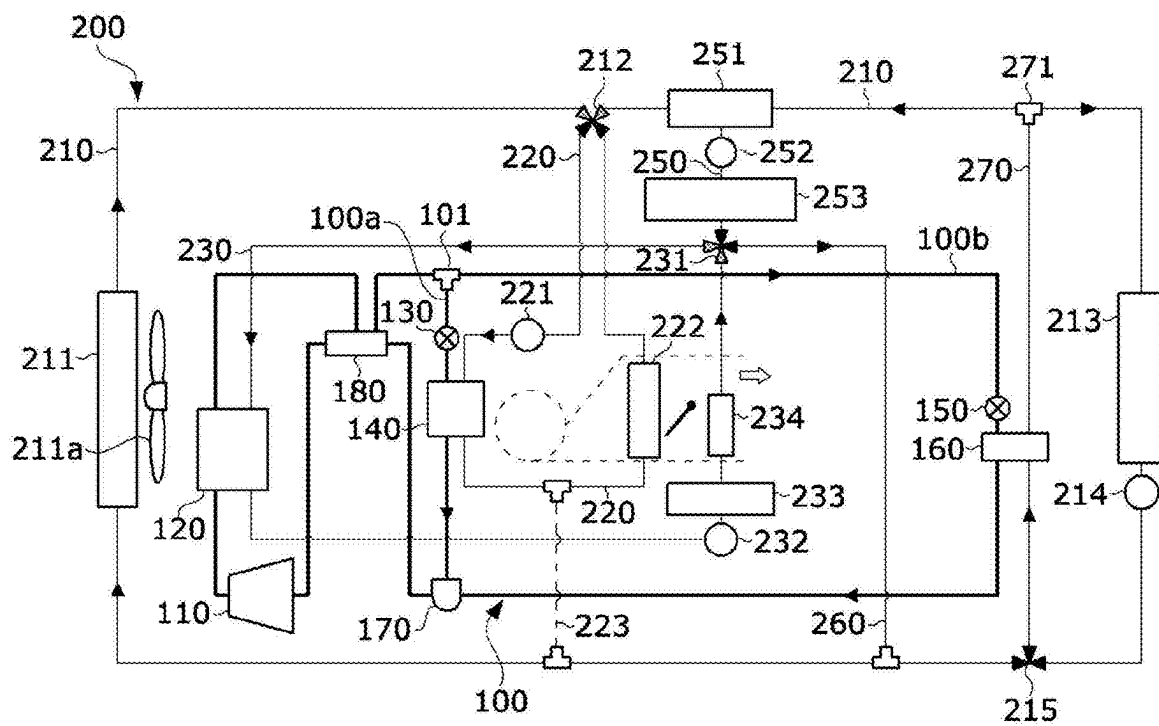


FIG. 13

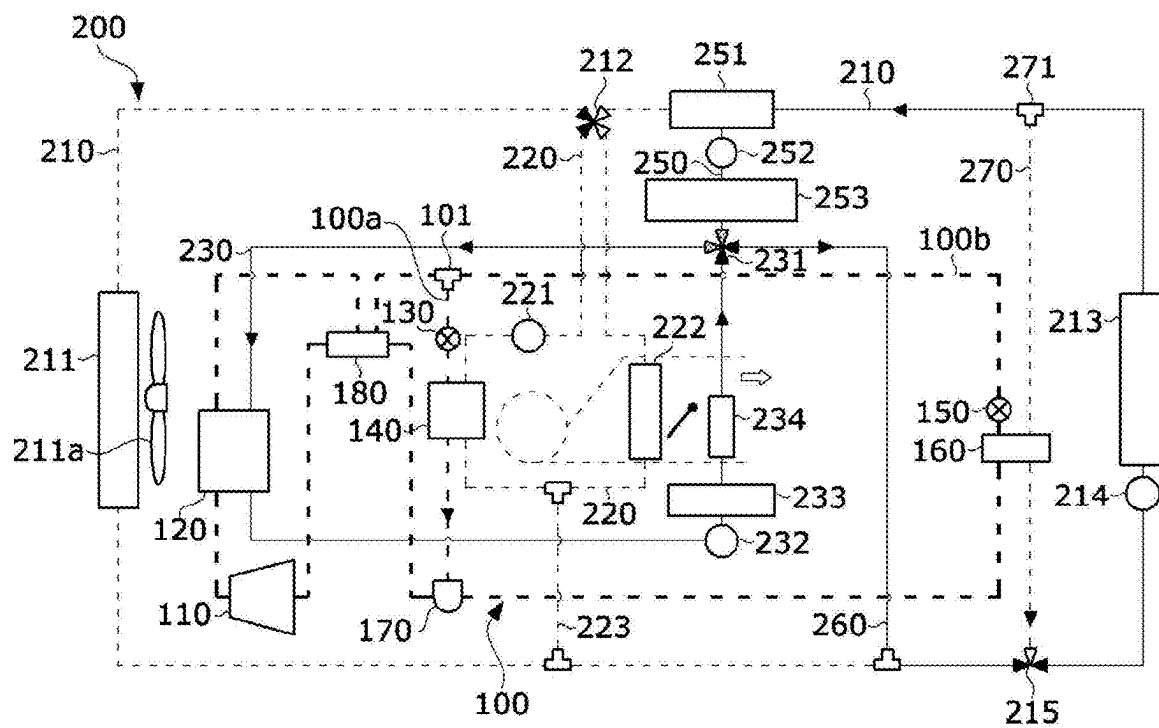


FIG. 14

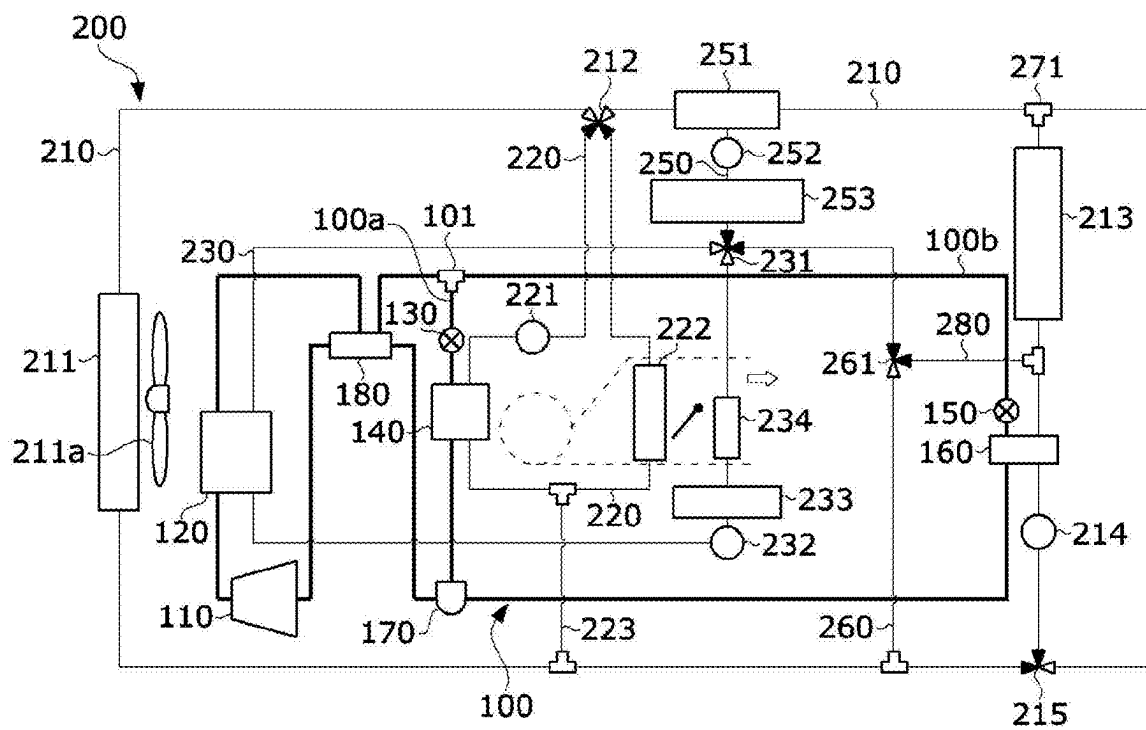


FIG. 15

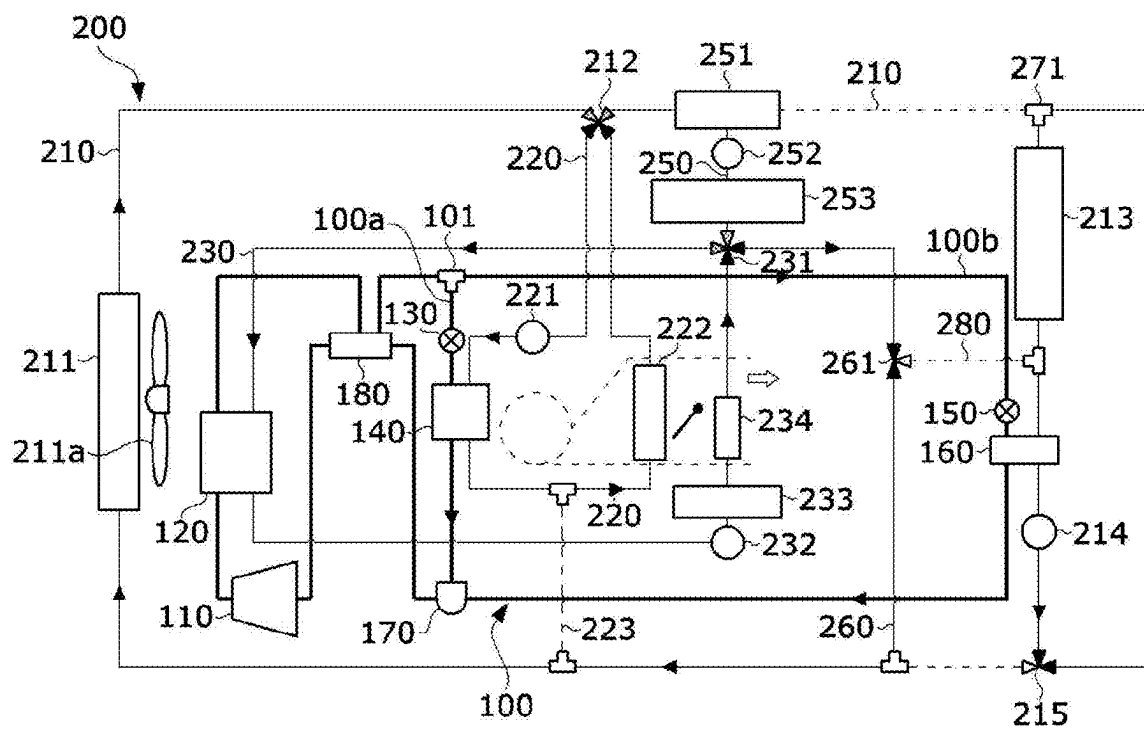


FIG. 16

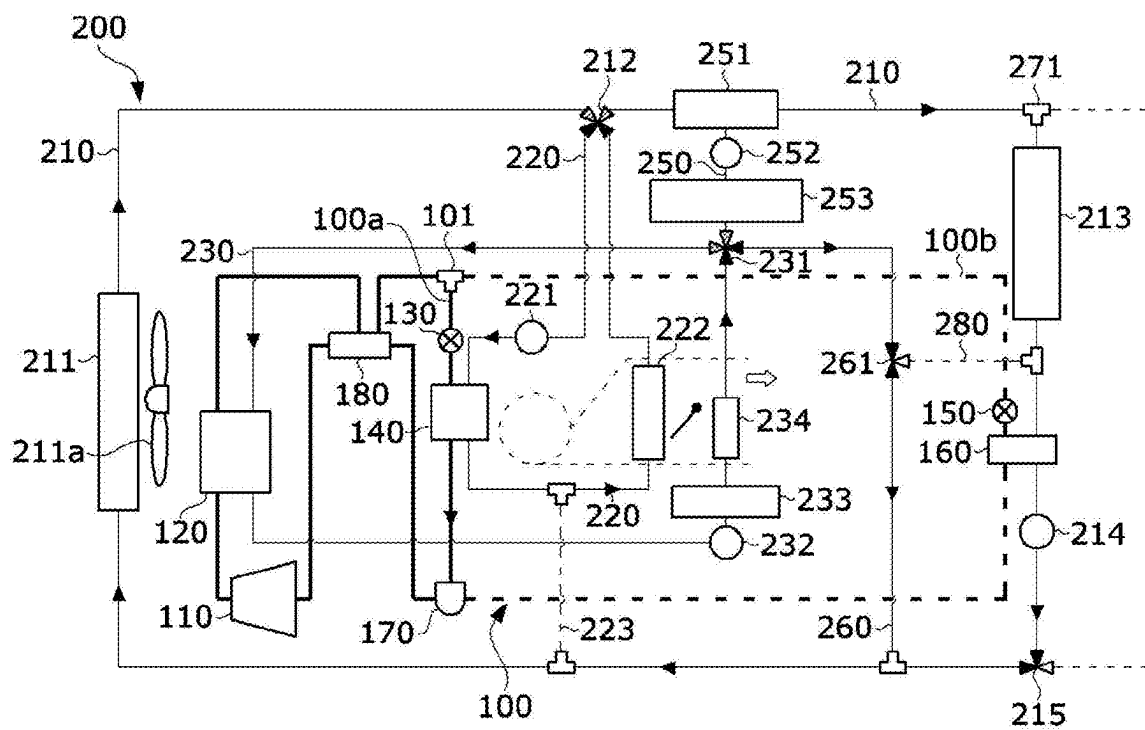


FIG. 17

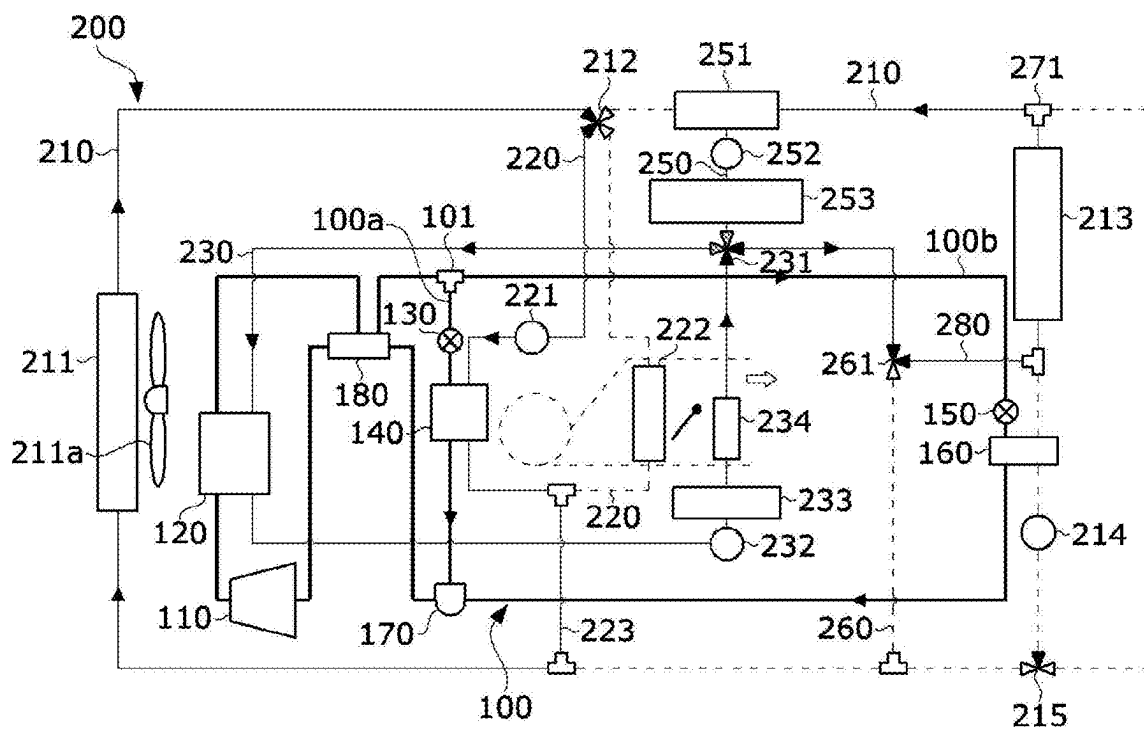


FIG. 18

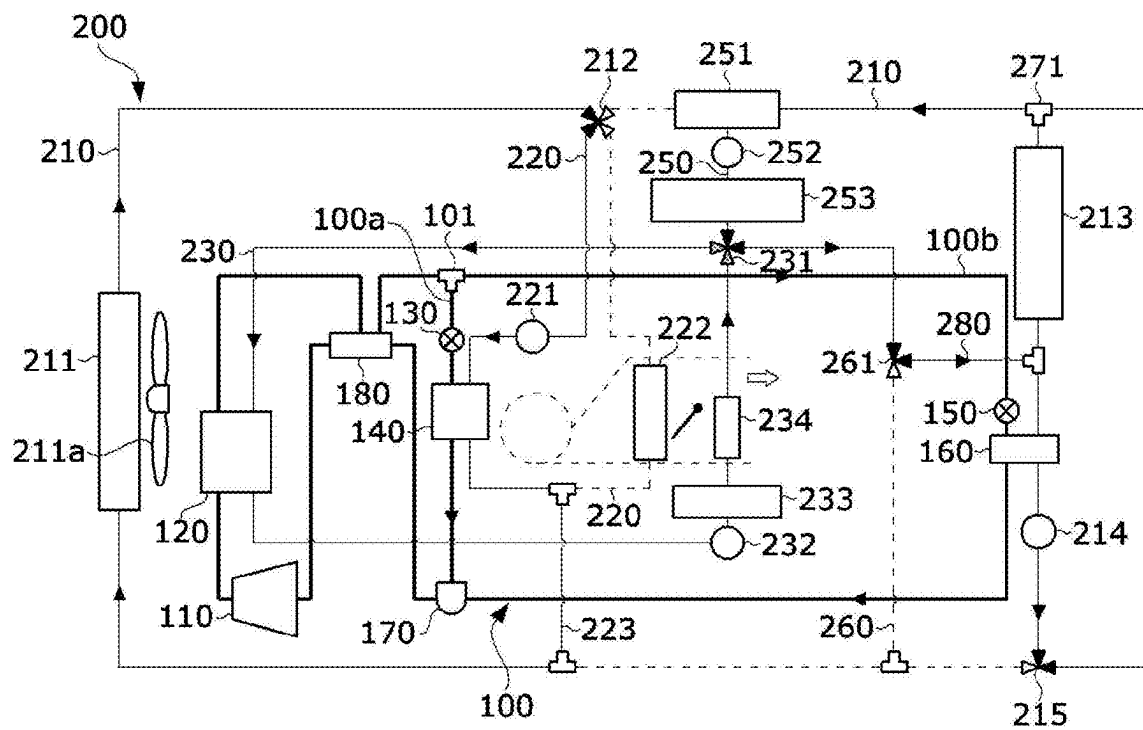


FIG. 19

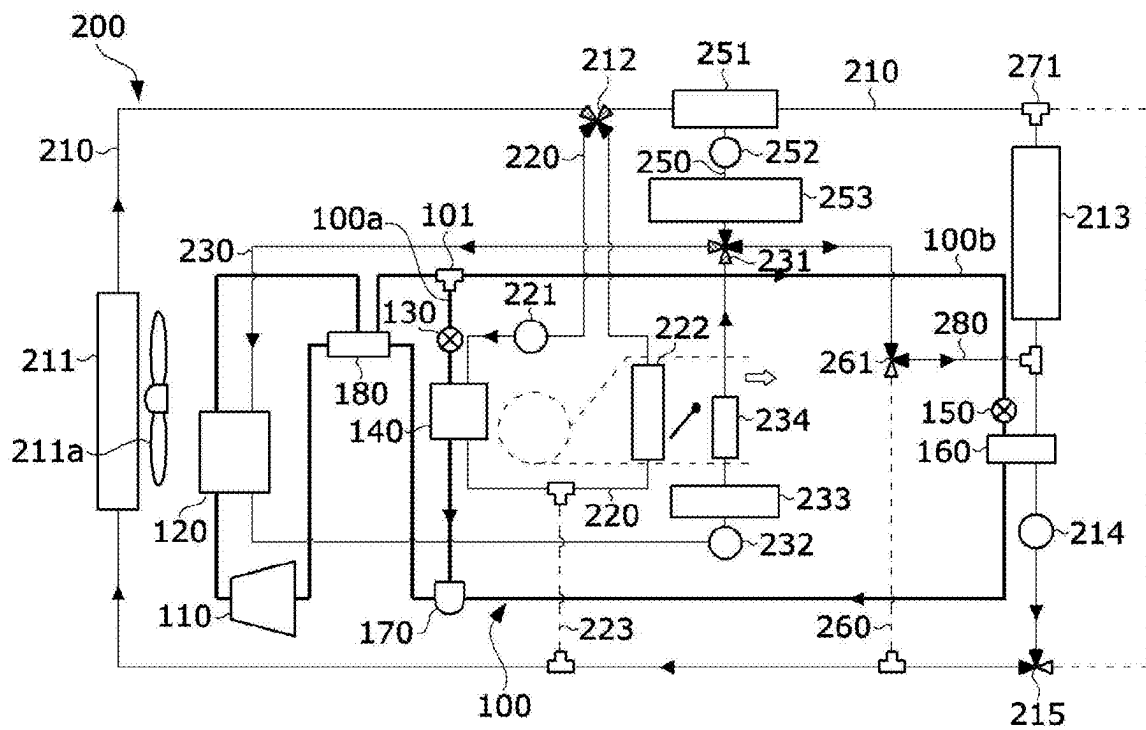


FIG. 20

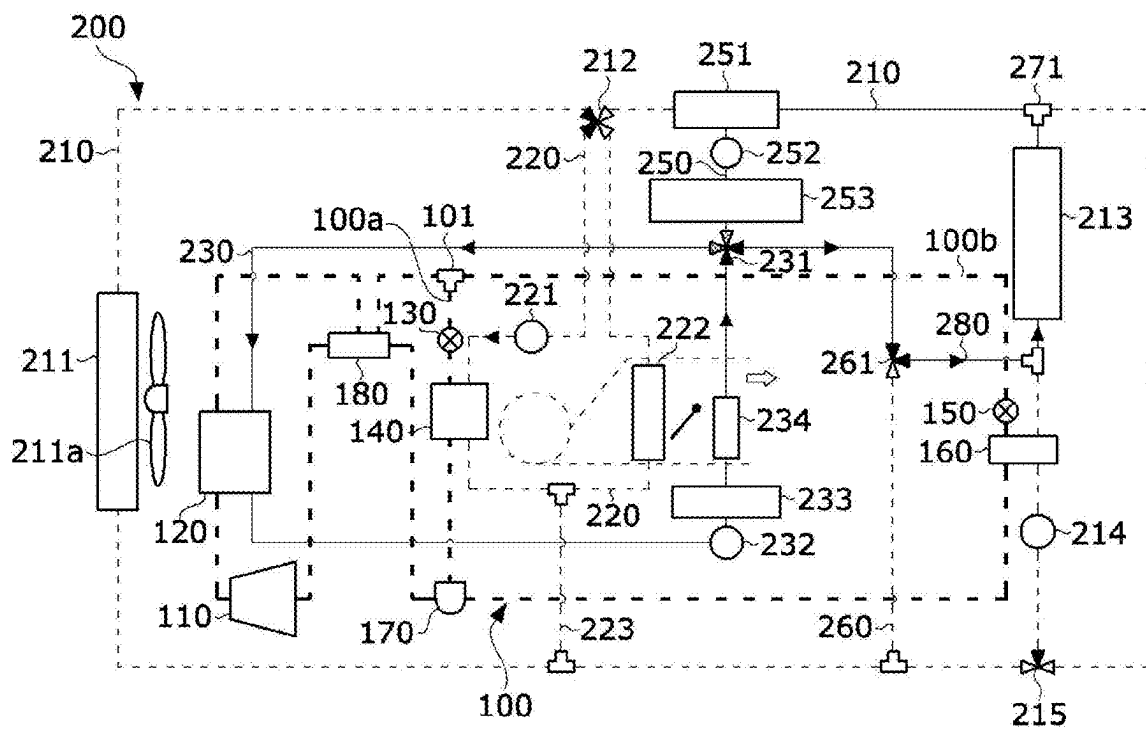


FIG. 21

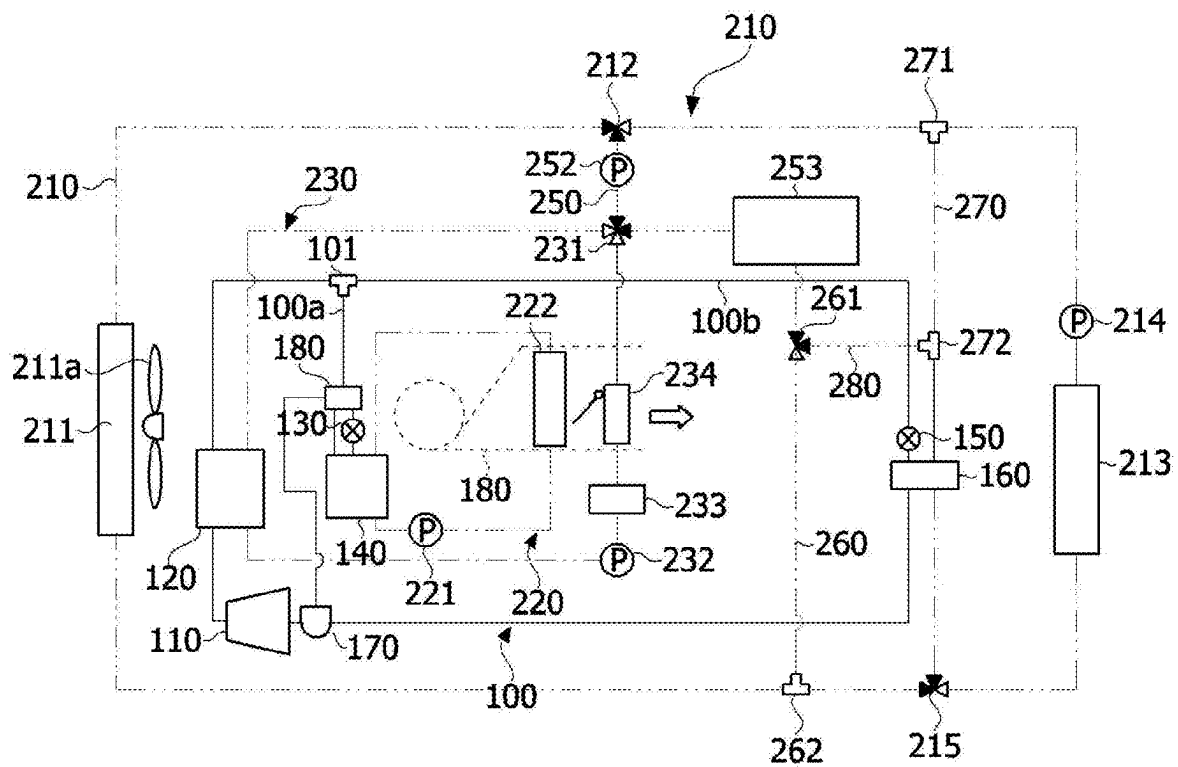


FIG. 22

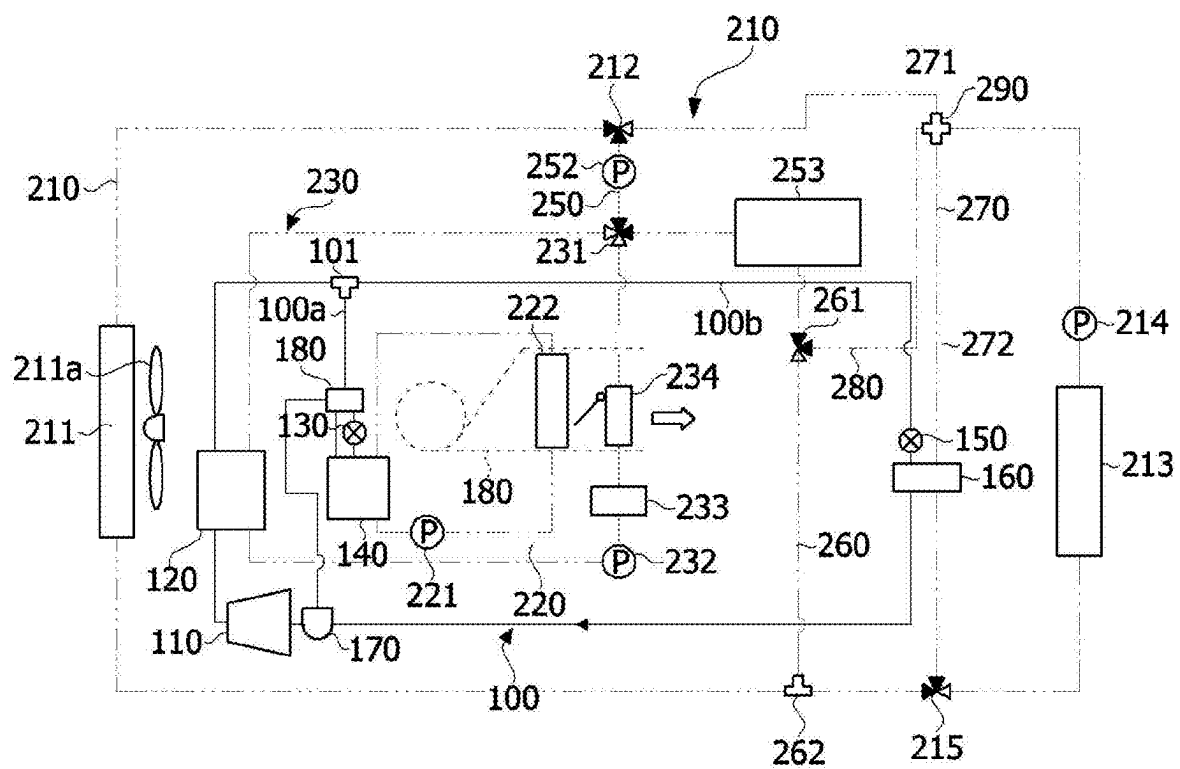


FIG. 23

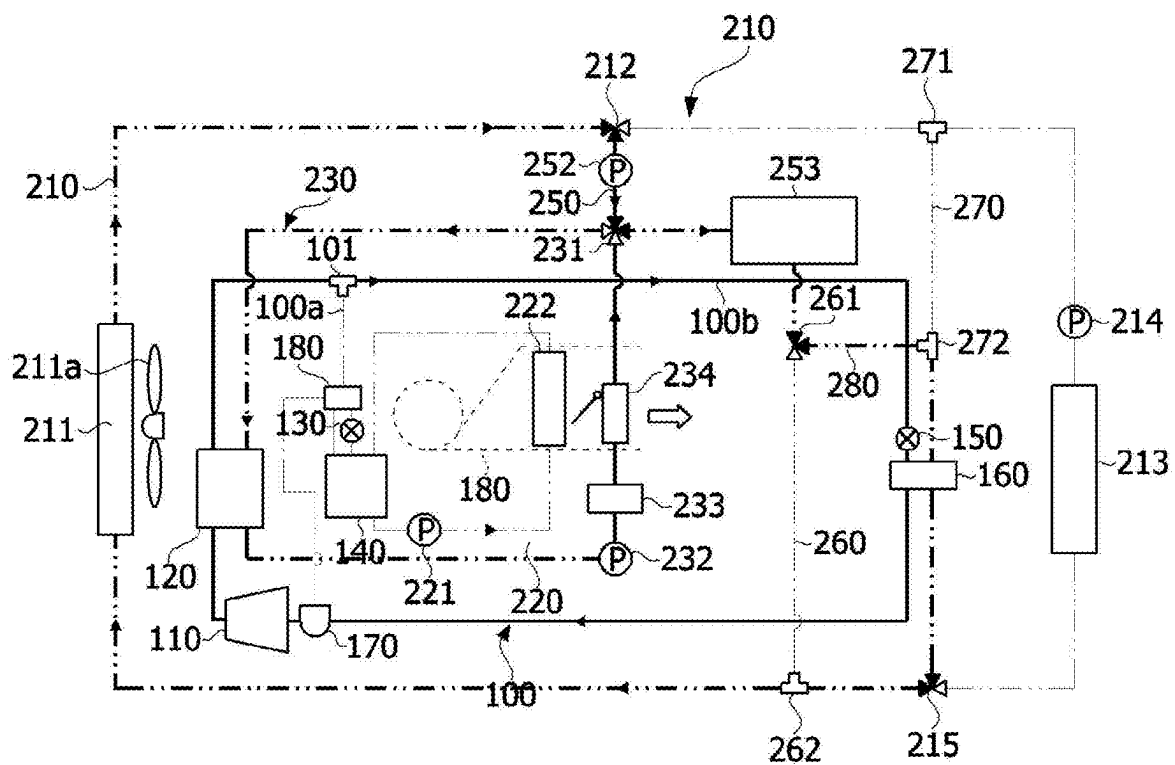


FIG. 24

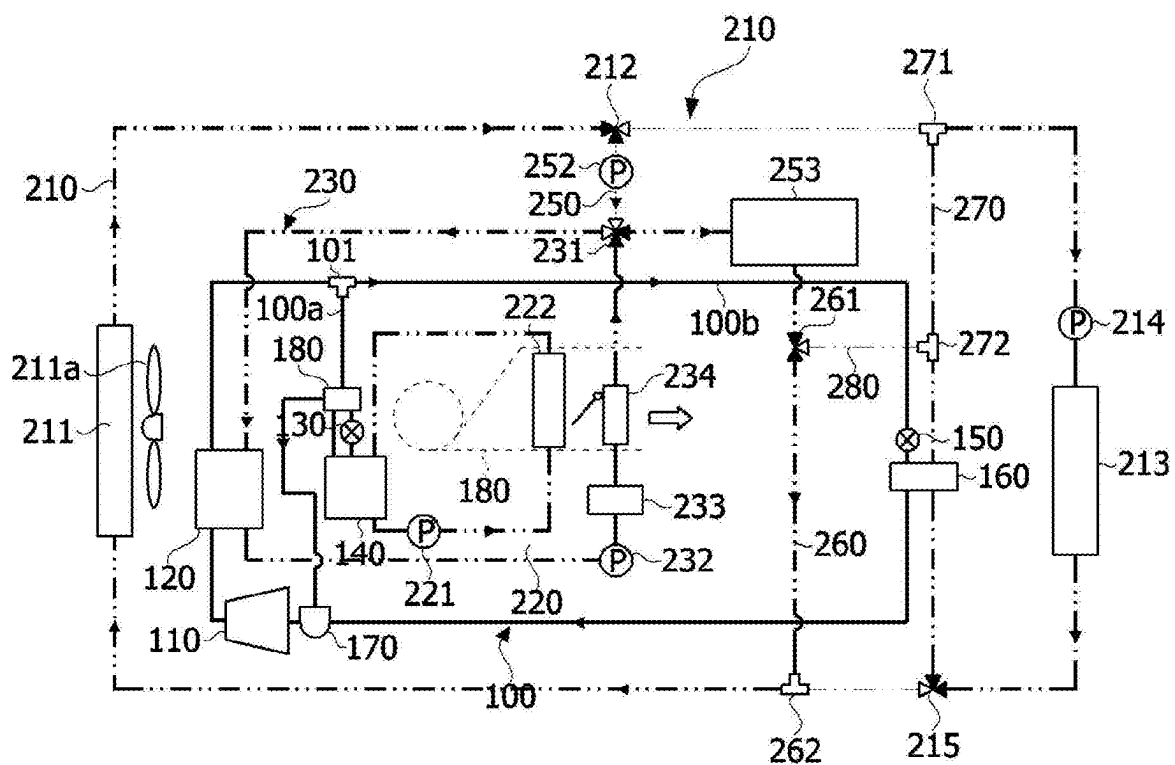
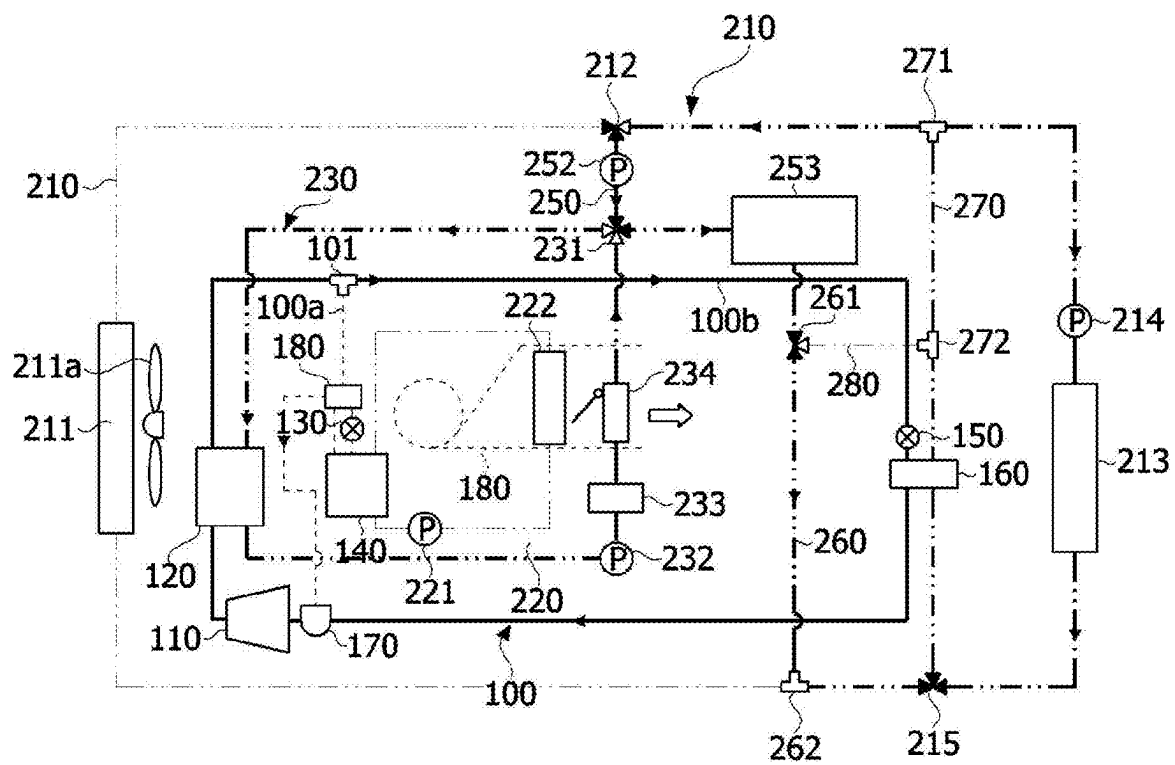


FIG. 25



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VEHICLE HEAT PUMP SYSTEM**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a national phase under 35 U.S.C. § 371 of International Application No. PCT/KR2020/017353 filed on Dec. 1, 2020, which claims the benefit of priority from Korean Patent Application Nos. 10-2019-0160086 filed on Dec. 4, 2019, 10-2019-0160087 filed on Dec. 4, 2019 and 10-2020-0157731 filed on Nov. 23, 2020. The entire contents of these applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

An embodiment relates to a vehicle heat pump system.

BACKGROUND ART

As development and research have been conducted on environmental-friendly technologies and alternative energy sources for replacing fossil raw materials, and electric vehicles and hybrid vehicles have been considered as most attractive fields in recent vehicle industries. Batteries are mounted in the electric vehicles and hybrid vehicles to provide driving power. The power of the battery is used not only to drive the vehicle, but also to cool or heat a vehicle interior.

When the battery is used as a heat source for cooling or heating the interior of the vehicle that provides driving power by using the battery, the traveling distance decreases to that extent. To solve this problem, a method of applying a heat pump system, which has been widely used as a domestic cooling or heating device in the related art, to the vehicle has been proposed.

For reference, the heat pump refers to a process of absorbing low-temperature heat and transferring the absorbed heat to a high-temperature location. For example, the heat pump implements a cycle in which a liquid refrigerant becomes a gaseous refrigerant by evaporating in an evaporator and absorbing heat from the surrounding, and the gaseous refrigerant becomes the liquid refrigerant by dissipating heat to the surrounding by means of a condenser. The application of the heat pump to the electric or hybrid vehicle may advantageously ensure an insufficient heat source in a general air conditioning casing in the related art.

FIG. 1 is a view illustrating a structure of a secondary loop cooling system of a vehicle heat pump system in the related art.

Referring to FIG. 1, a vehicle heat pump system in the related art performs cooling and heating by using a refrigerant circulation line circulating through a compressor 10, a condenser 20, an expansion valve 30, and a chiller 40, and a coolant circulation line that exchanges heat with a refrigerant passing through the chiller 40.

However, there is a problem in that the secondary loop cooling system has lower cooling performance than a direct cooling system. Because the secondary loop cooling system primarily cools a coolant and then cools a vehicle interior by cooling air with the coolant without directly cooling the air, there is a problem in that efficiency deteriorates.

DISCLOSURE**Technical Problem**

An object of the embodiment is to improve heating performance by absorbing heat of outside air in an evaporator by means of a radiator.

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Another object of the embodiment is to improve heating performance by selectively using outside air, waste heat of an electrical component, and waste heat of a battery as heat absorbing sources.

Still another object of the embodiment is to simplify a refrigerant circuit by using a secondary loop and ensure price competitiveness by reducing the number of heat exchangers and refrigerant valves.

Yet another object of the embodiment is to configure a coolant circulation line in which a coolant may flow through a radiator and an electrical component in series.

Still yet another object of the embodiment is to improve heating performance by allowing a coolant to flow at a constant flow rate in a coolant circulation line by using the structure.

Technical problems to be solved by the present invention are not limited to the above-mentioned technical problems, and other technical problems, which are not mentioned above, may be clearly understood by those skilled in the art from the following descriptions.

Technical Solution

An embodiment of the present invention provides a vehicle heat pump system including: a compressor configured to compress and circulate a refrigerant; a condenser configured to condense the compressed refrigerant; a first expansion valve configured to expand the condensed refrigerant; an evaporator configured to evaporate the refrigerant expanded by the first expansion valve by allowing the refrigerant to exchange heat with a coolant; a cabin cooler configured to cool a vehicle interior by allowing the coolant having passed through the evaporator to exchange heat with air; and a radiator configured to allow the coolant for cooling an electrical component to exchange heat with outside air, in which the coolant having passed through the radiator flows into the evaporator in a heating mode.

In particular, the vehicle heat pump system may further include: a heating line configured to heat the vehicle interior by circulating the coolant that exchanges heat with the refrigerant through the condenser; and a refrigerating line configured to cool a battery and the electrical component by circulating the coolant that exchanges heat with air or the refrigerant.

In particular, a fourth coolant joint and a second coolant joint may be disposed in the refrigerating line, such that a fifth connection line circulating through the radiator and the evaporator is defined.

In particular, the fifth connection line may be connected to a cooling line configured to cool the vehicle interior by circulating the coolant that exchanges heat with the refrigerant through the evaporator, and the cabin cooler may be disposed in the cooling line.

In particular, a third coolant joint configured to connect the cooling line may be disposed in the fifth connection line, and a second direction switching valve configured to determine whether to circulate the coolant to the cooling line may be disposed in the fifth connection line.

In particular, the cooling line may include a third pump connected between the evaporator and the cabin cooler, and the third pump may be disposed between the third coolant joint and the evaporator.

In particular, the vehicle heat pump system may include: a first connection line branching off from one side of the refrigerating line and connected to the heating line; and a second connection line branching off from the other side of the refrigerating line and connected to the heating line.

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In particular, the first connection line, the second connection line, and the heating line may be connected to a first direction switching valve, and the refrigerating line and the heating line may be connected to or disconnected from each other by the first direction switching valve.

In particular, the electrical component may be disposed in the second connection line, and a third connection line branching off from a fifth coolant joint disposed in the refrigerating line may be disposed to pass through the chiller.

In particular, a third direction switching valve may be disposed in the second connection line and control a flow direction of the coolant having passed through the electrical component, the third direction switching valve may be connected to a fourth coolant joint of the third connection line through a fourth connection line, and the fourth coolant joint may be disposed at a side adjacent to an inlet of the chiller.

In particular, the refrigerant having passed through the condenser may include a first refrigerant line branching off from one side of the refrigerant branch part and passing through the first expansion valve and the evaporator, and a second refrigerant line branching off from the other side of the refrigerant branch part and passing through the second expansion valve and the chiller.

In particular, a refrigerant heat exchanger may be disposed in the first refrigerant line, the refrigerant heat exchanger may be disposed between the refrigerant branch part and the first expansion valve, and the refrigerant introduced into the first expansion valve and the refrigerant having passed through the evaporator may exchange heat with each other.

In particular, the heating line may include: a heater core configured to heat the vehicle interior by using air heated as the air introduced into the vehicle interior exchanges heat with the coolant that exchanges heat with the refrigerant through the condenser; and a coolant heater disposed forward of the heater core based on the flow direction of the coolant and configured to heat the coolant.

In particular, in the heating mode, the second direction switching valve may close a flow path of the cooling line.

In particular, the third direction switching valve controls the direction of the coolant so that the coolant, which absorbs heat from the radiator and the electrical component, passes through the chiller.

In particular, in the cooling mode, the second direction switching valve may circulate the cooling line in a closed loop.

In particular, the vehicle heat pump system may include: a refrigerant circulation line in which the compressor, the condenser, the evaporator, and a chiller are disposed and the refrigerant circulates; a refrigerating line configured to cool a battery by circulating the coolant that exchanges heat with air or the refrigerant; a cooling line in which the cabin cooler is disposed, the cooling line being configured to cool the vehicle interior by using the coolant passing through the evaporator; and a cooling connection line configured to connect the cooling line and the refrigerating line, the refrigerating line and the cooling line may be connected by a fourth direction switching valve, and the refrigerating line and the cooling line may be separated and connected depending on an air conditioning mode.

In particular, one side of the cooling connection line may be disposed between the evaporator and the cabin cooler.

In particular, the refrigerant circulation line may include the compressor, the condenser, the refrigerant branch part, a first refrigerant line branching off from the refrigerant

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branch part toward one side and configured such that the first expansion valve and the evaporator are disposed therein, a second refrigerant line branching off from the refrigerant branch part toward the other side and configured such that the second expansion valve and the chiller are disposed therein, and an accumulator through which the refrigerant having passed through the first refrigerant line and the second refrigerant line passes.

In particular, the refrigerant circulation line may further include a refrigerant heat exchanger in which the refrigerant passing through the condenser and the refrigerant passing through the accumulator exchange heat with each other.

In particular, the vehicle heat pump system may further include a heating line configured to heat the vehicle interior by circulating the coolant that exchanges heat with the refrigerant through the condenser.

In particular, the vehicle heat pump system may include: a first connection line branching off from one side of the refrigerating line and connected to the heating line; and a second connection line branching off from the other side of the refrigerating line and connected to the heating line.

In particular, the first connection line, the second connection line, and the heating line may be connected to a first direction switching valve, and the refrigerating line and the heating line may be connected to or disconnected from each other by the first direction switching valve.

In particular, the electrical component may be disposed in the first connection line.

In particular, a third connection line branching off from a fifth coolant joint disposed in the refrigerating line may be disposed to pass through the chiller, and a fifth direction switching valve connected to the refrigerating line may be disposed in the third connection line.

In particular, the third connection line may be disposed in parallel with the refrigerating line in which the battery is disposed.

In particular, a reservoir tank may be disposed at a connection portion between the first connection line and the refrigerating line.

In particular, when the air conditioning mode is the heating mode, one region of the refrigerating line and one region of the cooling line may be connected by the operation of the fourth direction switching valve, such that the coolant may circulate.

In particular, the battery may be disposed in the third connection line, a fourth connection line may be disposed between the battery and the chiller, one side of the fourth connection line may be connected to the second connection line, the other side of the fourth connection line may be connected to the third connection line, and a third direction switching valve may be disposed in a region in which the second connection line and the fourth connection line are connected.

In particular, the vehicle heat pump system may include: a second expansion valve configured to expand the condensed refrigerant; a chiller configured to allow the refrigerant expanded by the second expansion valve to exchange heat with the coolant; and the electrical component configured to be cooled by the coolant, and in the heating mode, the coolant having passed through the radiator may exchange heat with the electrical component and then flow into the chiller.

In particular, the vehicle heat pump system may further include: a heating line configured to heat the vehicle interior by circulating the coolant that exchanges heat with the refrigerant through a first heat exchanger; and a refrigerating line configured to cool a heat generating component and the

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electrical component by circulating the coolant that exchanges heat with air or the refrigerant.

In particular, the vehicle heat pump system may include: a first connection line branching off from one side of the refrigerating line and connected to the heating line; and a second connection line branching off from the other side of the refrigerating line and connected to the heating line.

In particular, the first connection line, the second connection line, and the heating line may be connected to a first direction switching valve, and the refrigerating line and the heating line may be connected to or disconnected from each other by the first direction switching valve.

In particular, the electrical component may be disposed in the second connection line, and a third connection line branching off from a fifth coolant joint disposed in the refrigerating line may be disposed to pass through the chiller.

In particular, a third direction switching valve may be disposed in the second connection line and control a flow direction of the coolant having passed through the electrical component, the third direction switching valve may be connected to a fourth coolant joint of the third connection line through a fourth connection line, and the fourth coolant joint may be disposed at a side adjacent to an inlet of the chiller.

In particular, the fifth coolant joint and the fourth coolant joint may be integrated and have four branch lines.

In particular, the vehicle heat pump system may further include a second cooling line configured to cool the vehicle interior by circulating the coolant that exchanges heat with the refrigerant through the second heat exchanger.

In particular, the second cooling line may include: a cabin cooler configured to cool the vehicle interior by using air cooled as the air introduced into the vehicle interior exchanges heat with the refrigerant through the second heat exchanger; and a third pump connected between the second heat exchanger and the cabin cooler, and a coolant line connected to the second heat exchanger, the third pump, and the cabin cooler may define a closed loop.

In particular, the vehicle heat pump system may further include an air conditioning device having an air blower configured to blow air into the vehicle interior, the second heat exchanger and the third pump may be disposed outside the air conditioning device, and the cabin cooler may be disposed inside the air conditioning device.

In particular, the refrigerant having passed through the first heat exchanger may include a first refrigerant line branching off from one side of the refrigerant branch part and passing through the first expansion valve and the second heat exchanger, and a second refrigerant line branching off from the other side of the refrigerant branch part and passing through the second expansion valve and the chiller.

In particular, a refrigerant heat exchanger may be disposed in the first refrigerant line, the refrigerant heat exchanger may be disposed between the refrigerant branch part and the first expansion valve, and the refrigerant introduced into the first expansion valve and the refrigerant having passed through the second heat exchanger may exchange heat with each other.

In particular, the heating line may include: a heater core configured to heat the vehicle interior by using air heated as the air introduced into the vehicle interior exchanges heat with the coolant that exchanges heat with the refrigerant through the first heat exchanger; and a coolant heater disposed forward of the heater core based on the flow direction of the coolant and configured to heat the coolant.

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In particular, in the cooling mode, the heating line and the refrigerating line may be connected to each other.

In particular, the third direction switching valve may block the flow of the coolant into the chiller.

In particular, in the heating mode, the heating line and the refrigerating line may be separated from each other, and the coolant flowing along the refrigerating line may absorb heat while passing through the radiator and the electrical component and exchange heat with the refrigerant in the chiller.

Advantageous Effects

According to the embodiment, in the evaporator, the radiator may absorb heat of outside air, thereby improving heating performance.

In addition, the flow rate in the heat-absorbing cooling flow path may be increased by changing the position of the pump for cooling the vehicle interior, thereby improving heating performance.

In addition, outside air, waste heat of the electrical component, and waste heat of the battery may be selectively used as a heat absorbing source, thereby improving the heating performance according to the heating situation.

According to the embodiment, it is possible to simplify the refrigerant circuit by using the secondary loop and ensure price competitiveness by reducing the number of heat exchangers and refrigerant valves.

In addition, it is possible to implement various air conditioning modes by using the plurality of valve structures.

In addition, it is possible to heat the vehicle interior only by using waste heat of the electrical component and waste heat of the battery.

In addition, it is possible to increase the temperature of the battery while heating the vehicle interior.

In addition, it is possible to improve the heat-absorbing performance and heating performance by improving the structure of the system.

In addition, it is possible to improve the heating efficiency while implementing the cooling and heating circulation lines in the related art by improving the structure.

In addition, it is possible to improve the heating performance by improving the performance in absorbing waste heat of the electrical component. Further, it is possible to implement the maximum heating performance by using the structure capable of simultaneously absorbing heat of outside air and waste heat of the electrical component.

The various, beneficial advantages and effects of the present disclosure are not limited to the above-mentioned contents and may be more easily understood during the process of describing the specific embodiments of the present disclosure.

DESCRIPTION OF DRAWINGS

FIG. 1 is a view illustrating a structure of a secondary loop cooling system of a vehicle heat pump system in the related art according to a first embodiment of the present invention.

FIG. 2 is a structural view of a vehicle heat pump system according to an embodiment of the present invention.

FIG. 3 is a view illustrating a first embodiment of FIG. 2.

FIG. 4 is a view illustrating an operating state of the system in a heating mode in FIG. 2.

FIG. 5 is a view illustrating an operating state of the system in a heating mode in FIG. 3.

FIG. 6 is a view illustrating an operating state of the system in a cooling mode in FIG. 3.

FIG. 7 is a structural view of a vehicle heat pump system according to a second embodiment of the present invention.

FIG. 8 is a view illustrating an operation in a first air conditioning mode in FIG. 7.

FIG. 9 is a view illustrating an operation in a second air conditioning mode in FIG. 7.

FIG. 10 is a view illustrating an operation in a third air conditioning mode in FIG. 7.

FIG. 11 is a view illustrating an operation in a fourth air conditioning mode in FIG. 7.

FIG. 12 is a view illustrating an operation in a fifth air conditioning mode in FIG. 7.

FIG. 13 is a view illustrating an operation in a sixth air conditioning mode in FIG. 1.

FIG. 14 is a structural view of a vehicle heat pump system according to a third embodiment of the present invention.

FIG. 15 is a view illustrating an operation in a first air conditioning mode in FIG. 14.

FIG. 16 is a view illustrating an operation in a second air conditioning mode in FIG. 14.

FIG. 17 is a view illustrating an operation in a third air conditioning mode in FIG. 14.

FIG. 18 is a view illustrating an operation in a fourth air conditioning mode in FIG. 14.

FIG. 19 is a view illustrating an operation in a fifth air conditioning mode in FIG. 14.

FIG. 20 is a view illustrating an operation in a sixth air conditioning mode in FIG. 8.

FIG. 21 is a structural view of a vehicle heat pump system according to a fourth embodiment of the present invention.

FIG. 22 is a view illustrating a first embodiment of FIG. 21.

FIG. 23 is a view illustrating an operating state of the system in a maximum heating mode in FIG. 21.

FIG. 24 is a view illustrating an operating state of the system in a cooling mode in FIG. 21.

FIG. 25 is a view illustrating an operating state of the system in a general heating mode.

MODE FOR INVENTION

Hereinafter, exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings.

However, the technical spirit of the present invention is not limited to some embodiments described herein but may be implemented in various different forms. One or more of the constituent elements in the embodiments may be selectively combined and substituted within the scope of the technical spirit of the present invention.

In addition, unless otherwise specifically and explicitly defined and stated, the terms (including technical and scientific terms) used in the embodiments of the present invention may be construed as the meaning which may be commonly understood by the person with ordinary skill in the art to which the present invention pertains. The meanings of the commonly used terms such as the terms defined in dictionaries may be interpreted in consideration of the contextual meanings of the related technology.

In addition, the terms used in the embodiments of the present invention are for explaining the embodiments, not for limiting the present invention.

In the present specification, unless particularly stated otherwise, a singular form may also include a plural form. The expression “at least one (or one or more) of A, B, and C” may include one or more of all combinations that can be made by combining A, B, and C.

In addition, the terms first, second, A, B, (a), and (b) may be used to describe constituent elements of the embodiments of the present invention.

These terms are used only for the purpose of discriminating one constituent element from another constituent element, and the nature, the sequences, or the orders of the constituent elements are not limited by the terms.

Further, when one constituent element is described as being ‘connected’, ‘coupled’, or ‘attached’ to another constituent element, one constituent element can be connected, coupled, or attached directly to another constituent element or connected, coupled, or attached to another constituent element through still another constituent element interposed therebetween.

In addition, the explanation “one constituent element is formed or disposed above (on) or below (under) another constituent element” includes not only a case in which the two constituent elements are in direct contact with each other, but also a case in which one or more additional constituent elements are formed or disposed between the two constituent elements. In addition, the expression “above (on) or below (under)” may include a meaning of a downward direction as well as an upward direction based on one constituent element.

Hereinafter, embodiments will be described in detail with reference to the accompanying drawings. The same or corresponding constituent elements are assigned with the same reference numerals regardless of reference numerals, and the repetitive description thereof will be omitted.

FIGS. 2 to 25 clearly illustrate only main features for conceptually and clearly understanding the present invention. As a result, various modifications of the drawings are expected, and the scope of the present invention need not be limited to particular shapes illustrated in the drawings.

FIGS. 2 to 6 illustrate a first embodiment of the present invention.

FIG. 2 is a structural view of a vehicle heat pump system according to an embodiment of the present invention.

Referring to FIG. 2, the vehicle heat pump system according to the embodiment of the present invention may include a refrigerant circulation line 100 configured to circulate a refrigerant and cool a vehicle interior; and a coolant circulation line 200 configured to circulate a coolant, heat the vehicle interior, and cool components. Further, the coolant circulation line 200 may include a heating line 230 configured to heat the vehicle interior, and a refrigerating line 210 configured to cool an electrical component 253 and a battery 213.

The refrigerant circulation line 100 may include a compressor 110, a condenser 120, a refrigerant branch part 101, a first refrigerant line 100a branching off from the refrigerant branch part 101 toward one side and configured to pass through a first expansion valve and an evaporator 140, a second refrigerant line 100b branching off from the refrigerant branch part 101 toward the other side and configured to pass through a second expansion valve and a chiller 160, and an accumulator 170 through which the refrigerant having passed through the first refrigerant line 100a and the second refrigerant line 100b passes. The refrigerant having passed through the accumulator 170 flows into the compressor 110 again, such that refrigerant circulation is implemented.

In this case, a refrigerant heat exchanger 180 may be disposed in the first refrigerant line 100a. The refrigerant heat exchanger 180 may be disposed between the refrigerant branch part 101 and the first expansion valve 130, such that

the refrigerant introduced into the first expansion valve **130** may exchange heat with the refrigerant having passed through the evaporator **140**.

The compressor **110** operates by receiving power from an engine (internal combustion engine) or a motor. The compressor **110** sucks the refrigerant, compresses the refrigerant into a high-temperature, high-pressure gaseous refrigerant, and then discharges the refrigerant to the condenser **120**.

The condenser **120** serves as a condenser in both a cooling mode and a heating mode. The refrigerant flowing through the condenser **120** exchanges heat with the coolant in the coolant circulation line **200** to be described below and then flows to the first expansion valve **130**. As described above, the coolant heated by the refrigerant in the condenser **120** may be supplied to an interior heat exchanger through the coolant circulation line **200**. In one embodiment, a water-cooled condenser **120** may be used as the condenser **120**.

The refrigerant branch part **101** distributes the refrigerant to the first refrigerant line **100a** and the second refrigerant line **100b**.

The first expansion valve **130** and the evaporator **140** may be disposed in the first refrigerant line **100a**.

The first expansion valve **130** may throttle the refrigerant introduced from the first refrigerant line **100a**, perform a bypass operation on the refrigerant, or block a flow of the refrigerant. The first expansion valve **130** may be disposed at a side adjacent to an inlet of the evaporator **140** based on a flow direction of the refrigerant.

The evaporator **140** is installed in an air conditioning casing and disposed in the first refrigerant line **100a**. During a process in which the refrigerant discharged from the first expansion valve **130** is supplied to the evaporator **140** and air flowing in the air conditioning casing by a blower passes through the evaporator **140**, the air exchanges heat with the low-temperature, low-pressure refrigerant in the evaporator **140** and is converted into cold air. The cold air is discharged into a vehicle interior and cools an occupant compartment.

The second expansion valve and the chiller **160** may be disposed in the second refrigerant line **100b**.

The second expansion valve may throttle the refrigerant introduced from the second refrigerant line **100b**, perform a bypass operation on the refrigerant, or block a flow of the refrigerant. The second expansion valve may be disposed at a side adjacent to an inlet of the chiller **160** based on the flow direction of the refrigerant.

The low-temperature, low-pressure refrigerant discharged from the second expansion valve is supplied to the chiller **160** and exchanges heat with the coolant flowing in the coolant circulation line **200**. The cold coolant made by heat exchange in the chiller **160** may circulate through the coolant circulation line **200** and exchange heat with the high-temperature battery **213**. That is, the battery **213** exchanges heat with the coolant instead of exchanging heat with the refrigerant.

The accumulator **170** is installed at a side adjacent to an inlet of the compressor **110**. The refrigerant having passed through the evaporator **140** and/or the chiller **160** merges into the accumulator **170**. The accumulator **170** may divide the refrigerant into a liquid refrigerant and a gaseous refrigerant and supply only the gaseous refrigerant to the compressor **110**.

The refrigerant heat exchanger **180** allows the refrigerant introduced into the first expansion valve **130** and the refrigerant discharged from the evaporator **140** to exchange heat with each other, thereby improving cooling performance. In this case, an inlet side refrigerant line configured to connect the refrigerant branch part **101** and the first expansion valve

130 and configured such that the refrigerant is introduced into the evaporator **140** therethrough passes through the refrigerant heat exchanger **180**, and an outlet side refrigerant line configured to connect the evaporator **140** and the accumulator **170** and configured such that the refrigerant is discharged from the evaporator **140** therethrough passes through the refrigerant heat exchanger **180**, such that the refrigerant passing through the inlet side refrigerant line and the refrigerant passing through the outlet side refrigerant line may exchange heat with each other.

Therefore, before the refrigerant is introduced into the first expansion valve **130**, the refrigerant may be further cooled by the refrigerant heat exchanger **180**, thereby improving cooling performance implemented by the evaporator **140** and improving efficiency of the cooling system. In particular, the refrigerant heat exchanger **180** and the chiller **160** are connected in parallel.

That is, the refrigerant heat exchanger **180** is not disposed in series between the condenser **120** and the chiller **160** in the refrigerant line, but the refrigerant heat exchanger **180** is disposed adjacent to the evaporator **140**, such that the refrigerant heat exchanger **180** and the evaporator **140** may be disposed and connected in series. If the refrigerant heat exchanger is disposed in series between the water-cooled condenser **120** and the chiller **160**, the heating performance may decrease because a pressure drop occurs at a low-pressure side in the heating mode. On the contrary, when the refrigerant heat exchanger is connected in parallel, both the cooling performance and the heating performance increase. This is because there is no refrigerant heat exchanger between the condenser **120** and the chiller **160** in the flow of the refrigerant in the heating mode.

The coolant circulation line **200** may include the heating line **230** configured to heat the vehicle interior, and the refrigerating line **210** and a cooling line **220** configured to cool the electrical component **253** and the battery **213**.

The heating line **230** may include the condenser **120**, a second pump **232**, a coolant heater **233**, a heater core **234**, and a first direction switching valve **231**.

As described above, the refrigerant and the coolant may exchange heat with each other while passing through the condenser **120**.

The second pump **232** is a means for pumping the coolant so that the coolant circulates along the heating line **230**. The second pump **232** may be installed in the coolant line and disposed rearward of the condenser **120** based on the flow direction of the coolant.

The coolant heater **233** refers to a device for heating the coolant. The coolant heater **233** is connected and disposed rearward of the second pump **232** and forward of the heater core **234** based on the flow direction of the coolant. Further, the coolant heater **233** may operate when a temperature of the coolant is equal to or lower than a particular temperature. Various components such as an induction heater, a sheath heater, a PTC heater, or a film heater capable of generating heat using electric power may be used as the coolant heater **233**.

The heater core **234** may be disposed in an air conditioning device **190** of the vehicle. The air flowing by the air blower may be heated while passing through the heater core **234**, supplied to the vehicle interior, and used to heat the vehicle interior. Further, the heater core **234** may be connected and disposed rearward of the coolant heater **233** based on the flow direction of the coolant.

The first direction switching valve **231** may be installed between the heater core **234** and the condenser **120** and

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configured to selectively connect or disconnect the heating line **230** and the refrigerating line **210** to be described below.

More specifically, the first direction switching valve **231** may be installed in the heating line **230**. Two coolant line pipes may be connected to the first direction switching valve **231**. A single first connection line **250** branching off from one side of the refrigerating line **210** may be connected to the first direction switching valve **231**. A single second connection line branching off from the other side of the refrigerating line **210** may be connected to the first direction switching valve **231**. That is, four coolant lines may be connected to the first direction switching valve **231** so as to meet together. The first direction switching valve **231** may be a 4-way direction switching valve capable of adjusting states in which the four coolant lines are connected to or disconnected from one another.

The refrigerating line **210** may include a radiator **211**, a fourth direction switching valve **212**, a fourth pump **252**, a first direction switching valve **231**, the electrical component **253**, a fifth coolant joint **271**, a sixth coolant joint **262**, a first pump **214**, the battery **213**, the chiller **160**, and a fifth direction switching valve **215**.

The radiator **211** cools the coolant having exchanged heat with the electrical component **253** or the battery **213**. The radiator **211** may be cooled by a cooling fan **211a** in an air-cooled manner.

The fourth direction switching valve **212** may be installed in the refrigerating line **210**. Two coolant pipes may be connected to the fourth direction switching valve **212**. The first direction switching valve **231** and the fourth direction switching valve **212** may be connected by the first connection line **250** so that the heating line **230** and the refrigerating line **210** are connected.

That is, three coolant lines may be connected to the fourth direction switching valve **212** so as to meet together. The fourth direction switching valve **212** may be a 3-way direction switching valve capable of adjusting states in which the three coolant lines are connected to or disconnected from one another.

The fourth pump **252** is a means for pumping the coolant so that the coolant circulates along the refrigerating line **210**. Further, the fourth pump **252** is installed in the first connection line **250** and disposed between the first direction switching valve **231** and the fourth direction switching valve **212**. The operation of the fourth pump **252** may allow the coolant to flow from the fourth direction switching valve **212** to the first direction switching valve **231**.

The first direction switching valve **231** is as described above with reference to the heating line **230**.

The electrical component **253** is disposed in a second connection line **260** that connects the first direction switching valve **231** and the sixth coolant joint **262**. The electrical component **253** may be cooled by the coolant. Further, the electrical component **253** may be a drive motor, an inverter, a charger (onboard charger (OBC)), or the like.

The first pump **214** is a means for pumping the coolant so that the coolant circulates along the refrigerating line **210**. Further, the first pump **214** is installed in the coolant line and disposed between the fifth coolant joint **271** and the battery **213**, such that the coolant may flow from the first pump **214** to the battery **213**.

The battery **213** serves as a power source for the vehicle. The battery **213** may serve as a driving source for various types of electrical components **253** in the vehicle. In addition, the battery **213** may be connected to a fuel cell and serve to store electricity. Alternatively, the battery **213** may serve to store electricity supplied from the outside. Further,

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the battery **213** may be disposed in the coolant line and provided between the first pump **214** and the fifth direction switching valve **215**. Therefore, the battery **213** may be cooled or heated by exchanging heat with the flowing coolant.

The fifth coolant joint **271** is installed in the coolant line and disposed rearward of the fourth direction switching valve **212** based on the flow direction of the coolant. Three coolant lines are connected to the fifth coolant joint **271** so as to meet together. That is, the fifth coolant joint **271** may be installed such that two opposite sides thereof are connected to the refrigerating line **210**, and a third connection line **270** may be connected to a lower side of the fifth coolant joint **271**. In this case, the third connection line **270** may be connected to pass through the chiller **160**.

The sixth coolant joint **262** may be installed at a point at which a rear end of the second connection line **260** meets the refrigerating line **210**. Three coolant lines are connected to the sixth coolant joint **262** so as to meet together. That is, the sixth coolant joint **262** may be installed such that two opposite sides thereof are connected to the refrigerating line **210**, and the second connection line **260** may be connected to an upper side of the sixth coolant joint **262**.

The chiller **160** is as described above with reference to the heating line **230**.

The fifth direction switching valve **215** may be installed in the coolant line and disposed between the battery **213** and the sixth coolant joint **262**. Two coolant pipes may be connected to the fifth direction switching valve **215**. The third connection line **270** may be connected to an upper side of the fifth direction switching valve **215**, such that the battery **213** and the third connection line **270** may be connected in parallel. In this case, the fourth direction switching valve **212** may be a 3-way direction switching valve capable of adjusting states in which the three coolant lines are connected to or disconnected from one another.

Further, the air blower may be installed at one side of the air conditioning device **190** to blow air. A temperature adjustment door may be installed in the air conditioning device **190**. In addition, the evaporator **140** and the heater core **234** disposed in the air conditioning device **190** may be disposed and configured such that on the basis of the operation of the temperature adjustment door, the air discharged from the air blower may flow into the vehicle interior while passing only through the evaporator **140** or flow into the vehicle interior while passing through the evaporator **140** and then passing through the heater core **234**.

The fifth connection line **290**, which corresponds to a first line of the refrigerating line, may be disposed in the refrigerating line **210**. The fifth connection line **290** may connect a first coolant joint **290c** and a second coolant joint **290b**, the first coolant joint **290c** may be disposed between the radiator **211** and the fourth direction switching valve **212**, and the second coolant joint **290b** may be disposed between the radiator **211** and the sixth coolant joint **262**.

The fifth connection line **290** may be disposed to pass through the evaporator **140**. The coolant having passed through the radiator **211** may circulate through the evaporator **140** according to the air conditioning mode.

In the fifth connection line **290**, a third coolant joint **290a** may be disposed at a side adjacent to an inlet of the evaporator **140**, and a second direction switching valve **291** may be disposed at a side adjacent to an outlet of the evaporator **140**.

The cooling line **220** may be connected to the third coolant joint **290a** and the second direction switching valve **291**.

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The cooling line 220 may include the evaporator 140, a cabin cooler 222, and a third pump 221. In this case, the coolant line, which connects the evaporator 140, the cabin cooler 222, and the third pump 221 may define a closed loop according to the operation of the second direction switching valve 291.

As described above, the refrigerant and the coolant may exchange heat with each other while passing through the evaporator 140.

The cabin cooler 222 serves as an air-cooled evaporator 140. The coolant cooled by exchanging heat with the refrigerant passing through the evaporator 140 passes through the cabin cooler 222. Further, the cabin cooler 222 is disposed in the air conditioning device 190. The air flowing by the air blower of the air conditioning device 190 is cooled while passing through the cabin cooler 222, supplied to the vehicle interior, and used to cool the vehicle interior.

The third pump 221 serves to pump the coolant so that the coolant circulates along the cooling line 220. Further, the third pump 221 is installed in the fifth connection line 290 and disposed between the evaporator 140 and the third coolant joint 290a. The operation of the third pump 221 may circulate the coolant.

Therefore, the vehicle heat pump system according to the present invention adopts the cooling line 220 using the coolant for cooling the vehicle interior, such that the refrigerant circulation line 100 through which the refrigerant circulates may be disposed outside the vehicle interior instead of being disposed in the vehicle interior. Therefore, it is possible to reduce a length of the refrigerant pipe, reduce the amount of refrigerant, and modularize the components in the refrigerant circulation line 100. In addition, a high-efficiency natural refrigerant may be used as the refrigerant used for the refrigerant circulation line 100, which may improve the efficiency of the thermal management system.

FIG. 3 is a view illustrating a first embodiment of FIG. 2. FIG. 3 is a view illustrating a structure in which a fourth connection line is disposed in FIG. 2.

Referring to FIG. 3, a third direction switching valve 261 may be disposed in the second connection line 260 and control a flow direction of the coolant having passed through the electrical component 253. The third direction switching valve 261 is disposed in the fourth connection line 280 branching off from the second connection line 260. The fourth connection line 280 may be connected to a fourth coolant joint 272 disposed in the third connection line 270.

The fourth coolant joint 272 is disposed at a side adjacent to an inlet of the chiller 160 through which the coolant is introduced. The coolant having passed through the electrical component 253 may be introduced into the chiller 160 according to the operation of the third direction switching valve 261.

In one embodiment, when the upper side and the right side of the third direction switching valve 261 are connected, the electrical component 253 and the chiller 160 may be connected in series.

FIG. 4 is a view illustrating an operating state of the system in a heating mode in FIG. 2.

Referring to FIG. 4, in the refrigerant circulation line 100, the compressor 110 operates, and the high-temperature, high-pressure refrigerant is discharged from the compressor 110. Further, the refrigerant discharged from the compressor 110 is cooled while exchanging heat with the coolant in the condenser 120. Thereafter, the refrigerant cooled and condensed in the condenser 120 is divided in the refrigerant branch part 101. A part of the refrigerant passes through the refrigerant heat exchanger 180 and then be throttled and

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expanded while passing through the first expansion valve 130. Thereafter, the expanded refrigerant exchanges heat with the coolant in a cooling line 220 while passing through the evaporator 140, and the coolant in the cooling line 220 is cooled by the refrigerant.

Further, the refrigerant evaporated in the evaporator 140 passes through the refrigerant heat exchanger 180, exchanges heat with the refrigerant before the refrigerant is introduced into the first expansion valve 130, and then flows into the compressor 110 again via the accumulator 170.

In addition, the remaining part of refrigerant divided in the refrigerant branch part 101 is throttled and expanded while passing through the second expansion valve 150. Thereafter, the expanded refrigerant is evaporated by exchanging heat with the coolant while passing through the chiller 160, such that the coolant may be cooled. Further, the refrigerant evaporated in the chiller 160 flows into the compressor 110 again via the accumulator 170.

As described above, the refrigerant having passed through the evaporator 140 and the refrigerant having passed through the chiller 160 merge with each other in the accumulator 170 and flow into the compressor 110. The refrigerant circulates as the above-mentioned process is repeated.

Meanwhile, the coolant in the coolant circulation line 200 is circulated by the operations of the second pump 232, the fourth pump 252, the first pump 214, and the third pump 221. Further, the coolant may be heated while passing through the condenser 120, heated by the coolant heater 212, and heated by waste heat of the electrical component 253. The coolant may be cooled while passing through the chiller 160. In this case, the first direction switching valve 231 and the fourth direction switching valve 212 may adjust the directions thereof so that the heating line 230 and the refrigerating line 210 are separated.

More specifically, the coolant may flow as the upper side and the right side of the first direction switching valve 231 are connected to each other, and the coolant may flow as the lower side and the left side of the first direction switching valve 231 are connected to each other. Further, when the right side and the lower side of the fourth direction switching valve 212 are connected to each other, the coolant may flow, and the left side of the fourth direction switching valve 212 may be disconnected.

In addition, all the upper side, the left side, and the right side of the fifth direction switching valve 215 may be opened.

Therefore, the coolant in the heating line 230 sequentially passes through the second pump 232, the coolant heater 212, the heater core 234, the first direction switching valve 231, and the condenser 120, flows into the second pump 232 again, and circulates. This cycle is repeated.

In this case, the coolant passes through the heater core 234 while exchanging heat with the air blown by the air blower of the air conditioning device 190, such that the air is heat. The heated air is supplied to the vehicle interior and used to heat the vehicle interior.

Further, the coolant in the refrigerating line 210 separated from the heating line 230 flows from the fourth pump 252 sequentially to the first direction switching valve 231, the electrical component 253, the sixth coolant joint 262, the fifth direction switching valve 215, the chiller 160, the fifth coolant joint 271, and the fourth direction switching valve 212, flows into the fourth pump 252 again, and circulates. This cycle is repeated.

In addition, the coolant having passed through the battery 213 may circulate through the first pump 214, merge into the

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fifth direction switching valve **215**, flows upward, and then be divided into two pieces in the fifth coolant joint **271**.

In this case, the left side of the fourth direction switching valve **212** is blocked, and a separate coolant circulation line may be formed by the third pump **221**.

The coolant having passed through the radiator **211** may be circulated downward from the first coolant joint **290c** by the third pump **221**, such that a circulation line in which the coolant passes through the evaporator **140** and the second coolant joint **290b** and then flows into the radiator **211** again may be defined. Therefore, in the evaporator **140**, the coolant may absorb heat from outside air by means of the radiator **211**. In addition, the third pump **221** may be disposed in the fifth connection line **290** to increase a flow rate in a heat absorbing coolant path, thereby improving heating performance.

In this case, the upper side and the lower side of the second direction switching valve **291** connected to the fifth connection line **290** are connected, and the right side of the second direction switching valve **291** is blocked, thereby preventing the coolant from circulating to the cooling line **220**.

FIG. **5** is a view illustrating an operating state of the system in a heating mode in FIG. **3**.

Referring to FIG. **5**, in the refrigerant circulation line **100**, the compressor **110** operates, and the high-temperature, high-pressure refrigerant is discharged from the compressor **110**. Further, the refrigerant discharged from the compressor **110** is cooled while exchanging heat with the coolant in the condenser **120**. Thereafter, the refrigerant cooled and condensed in the condenser **120** is divided in the refrigerant branch part **101**. A part of the refrigerant passes through the refrigerant heat exchanger **180** and then be throttled and expanded while passing through the first expansion valve **130**. Thereafter, the expanded refrigerant exchanges heat with the coolant in the cooling line **220** while passing through the evaporator **140**, and the coolant in the cooling line **220** is cooled by the refrigerant.

Further, the refrigerant evaporated in the evaporator **140** passes through the refrigerant heat exchanger **180**, exchanges heat with the refrigerant before the refrigerant is introduced into the first expansion valve **130**, and then flows into the compressor **110** again via the accumulator **170**.

In addition, the remaining part of refrigerant divided in the refrigerant branch part **101** is throttled and expanded while passing through the second expansion valve **150**. Thereafter, the expanded refrigerant is evaporated by exchanging heat with the coolant while passing through the chiller **160**, such that the coolant may be cooled. Further, the refrigerant evaporated in the chiller **160** flows into the compressor **110** again via the accumulator **170**.

As described above, the refrigerant having passed through the evaporator **140** and the refrigerant having passed through the chiller **160** merge with each other in the accumulator **170** and flow into the compressor **110**. The refrigerant circulates as the above-mentioned process is repeated.

The coolant in the coolant circulation line **200** is circulated by the operations of the second pump **232**, the fourth pump **252**, and the third pump **221**. Further, the coolant may be heated while passing through the condenser **120**, heated by the coolant heater **233**, and heated by waste heat of the electrical component **253**. The coolant may be cooled while passing through the chiller **160**.

In this case, the first direction switching valve **231** and the fourth direction switching valve **212** may adjust the directions thereof so that the heating line **230** and the refrigerating line **210** are separated. More specifically, the coolant may

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flow as the upper side and the right side of the first direction switching valve **231** are connected to each other, and the coolant may flow as the lower side and the left side of the first direction switching valve **231** are connected to each other. Further, when left side and the lower side of the fourth direction switching valve **212** are connected to each other, the coolant may flow, and the right side of the fourth direction switching valve **212** may be disconnected. In addition, the upper side and the left side of the fifth direction switching valve **215** may be connected to each other, and the right side of the fifth direction switching valve **215** may be blocked.

Therefore, the coolant in the heating line **230** sequentially passes through the second pump **232**, the coolant heater **233**, the heater core **234**, the first direction switching valve **231**, and the condenser **120**, flows into the second pump **232** again, and circulates. This cycle is repeated. In this case, the coolant passes through the heater core **234** while exchanging heat with the air blown by the air blower of the air conditioning device **190**, such that the air is heat. The heated air is supplied to the vehicle interior and used to heat the vehicle interior.

Further, the coolant in the refrigerating line **210** separated from the heating line **230** flows from the fourth pump **252** sequentially to the first direction switching valve **231**, the electrical component **253**, the third direction switching valve **261**, the fourth coolant joint **272**, the chiller **160**, the fifth direction switching valve **215**, and the radiator **211**. The coolant having passed through the radiator is divided in the fourth coolant joint **272**. A part of the coolant flows into the fourth pump **252** via the fourth direction switching valve **212**. The remaining part of the coolant flows along the fifth connection line **290**, performs heat exchange in the evaporator **140**, flows to the second coolant joint **290b** again, merges with the coolant passing through the fifth direction switching valve **215**, and then flows to the radiator **211**.

In this case, the upper side and the right side of the third direction switching valve **261** may be connected to each other so that the coolant flows to the chiller **160**. The coolant flows to the chiller **160** via the fourth coolant joint **272**.

Therefore, the electrical component **253** and the chiller **160** may be connected in series, and a coolant flow path capable of absorbing both outside air and waste heat may be formed. In this structure, since there is no branch part in the refrigerating line **210** circulating through the radiator **211**, the electrical component **253**, and the chiller **160**, the coolant may flow at an almost constant flow rate, which may improve the heating performance.

In this case, the coolant may not flow to the battery **213** through the fourth direction switching valve **212** and the fifth direction switching valve **215**.

FIG. **6** is a view illustrating an operating state of the system in a cooling mode in FIG. **3**.

Referring to FIG. **6**, in the refrigerant circulation line **100**, the compressor **110** operates, and the high-temperature, high-pressure refrigerant is discharged from the compressor **110**. Further, the refrigerant discharged from the compressor **110** is cooled while exchanging heat with the coolant in the condenser **120**. Thereafter, the refrigerant cooled and condensed in the water-cooled condenser **120** is divided in the refrigerant branch part **101**. A part of the refrigerant passes through the refrigerant heat exchanger **180** and then be throttled and expanded while passing through the first expansion valve **130**. Thereafter, the expanded refrigerant exchanges heat with the coolant in the cooling line **220** while passing through the evaporator **140**, and the coolant in the cooling line **220** is cooled by the refrigerant.

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Further, the refrigerant evaporated in the evaporator **140** passes through the refrigerant heat exchanger **180**, exchanges heat with the refrigerant before the refrigerant is introduced into the first expansion valve **130**, and then flows into the compressor **110** again via the accumulator **170**.

In addition, the remaining part of refrigerant divided in the refrigerant branch part **101** is throttled and expanded while passing through the second expansion valve. Thereafter, the expanded refrigerant is evaporated by exchanging heat with the coolant while passing through the chiller **160**, such that the coolant may be cooled. Further, the refrigerant evaporated in the chiller **160** flows into the compressor **110** again via the accumulator **170**.

As described above, the refrigerant having passed through the evaporator **140** and the refrigerant having passed through the chiller **160** merge with each other in the accumulator **170** and flow into the compressor **110**. The refrigerant circulates as the above-mentioned process is repeated.

In addition, the coolant is circulated in the cooling line **220** by the operation of the third pump **221**. Further, the coolant passes through the cabin cooler **222** while exchanging heat with the air blown by the air blower of the air conditioning device **190**, such that the air is cooled. The cooled air is supplied to the vehicle interior and used to cool the vehicle interior. In this case, the upper side and the right side of the second direction switching valve **291** disposed in the second connection line **290** are connected to each other, and the lower side of the second direction switching valve **291** is closed, such that the cooling line **220** circulates in a closed loop structure.

Meanwhile, the coolant in the coolant circulation line **200** is circulated by the operations of the second pump **232**, the fourth pump **252**, and the first pump **214**. Further, the battery **213**, the electrical component **253**, and the refrigerant passing through the condenser **120** may be cooled by the coolant. The heated coolant may be cooled while exchanging heat with the outside air by the operation of the cooling fan **211a** of the radiator **211** for the electrical component.

In this case, the first direction switching valve **231** and the fourth direction switching valve **212** may adjust the directions thereof so that the heating line **230** and the refrigerating line **210** are connected. More specifically, the coolant may flow as the upper side and the left side of the first direction switching valve **231** are connected to each other, and the coolant may flow as the lower side and the right side of the first direction switching valve **231** are connected to each other. Further, when the left side and the lower side of the fourth direction switching valve **212** are connected to each other, the coolant may flow, and the right side of the fourth direction switching valve **212** may be disconnected. In addition, the upper side and the right side of the fifth direction switching valve **215** may be connected to each other, and the left side of the fifth direction switching valve **215** may be blocked.

Therefore, the coolant flows from the radiator **211** sequentially to the fourth direction switching valve **212**, the fourth pump **252**, the first direction switching valve **231**, the condenser **120**, the second pump **232**, the coolant heater **233**, the heater core **234**, the first direction switching valve **231**, the electrical component **253**, and the sixth coolant joint **262**, flows into the radiator **211** again, and circulates. This cycle is repeated.

In this case, the right line of the third direction switching valve **261** may be closed, thereby preventing the coolant from flowing to the fourth connection line **280**.

The fourth direction switching valve **212** may prevent the coolant from flowing from the fourth direction switching

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valve **212** to the fifth coolant joint **271**, and the fifth direction switching valve **215** may prevent the coolant from flowing from the fifth direction switching valve **215** to the sixth coolant joint **262**.

In addition, the coolant may flow from the chiller **160** sequentially to the fifth coolant joint **271**, the first pump **214**, the battery **213**, and the fifth direction switching valve **215**, flows into the chiller **160** again, and circulates. This cycle is repeated. That is, the battery **213** and the chiller **160** may define a separate closed loop, in which the coolant circulates, in the refrigerating line **210** by the fourth direction switching valve **212** and the fifth direction switching valve **215**, such that the battery **213** may be separately cooled.

FIGS. 7 to 13 illustrate a second embodiment of the present invention.

FIG. 7 is a structural view of a vehicle heat pump system according to a second embodiment of the present invention, FIG. 8 is a view illustrating an operation in a first air conditioning mode in FIG. 7, FIG. 9 is a view illustrating an operation in a second air conditioning mode in FIG. 7, FIG. 10 is a view illustrating an operation in a third air conditioning mode in FIG. 7, FIG. 11 is a view illustrating an operation in a fourth air conditioning mode in FIG. 7, FIG. 12 is a view illustrating an operation in a fifth air conditioning mode in FIG. 7, and FIG. 13 is a view illustrating an operation in a sixth air conditioning mode in FIG. 1.

Referring to FIG. 7, the vehicle heat pump system according to the embodiment of the present invention may include the refrigerant circulation line **100** configured to circulate the refrigerant and cool the vehicle interior; and the coolant line **200** configured to circulate the coolant, heat the vehicle interior, and cool components. Further, the coolant line **200** may include the refrigerating line **210** and the heating line **230**.

The refrigerant circulation line **100** may include the compressor **110**, the water-cooled condenser **120**, the refrigerant branch part **101**, the first refrigerant line branching off from the refrigerant branch part **101** and configured such that the first expansion valve **130** and the water-cooled evaporator **140** are disposed therein, and the second refrigerant line branching off from the refrigerant branch part **101** and configured such that the second expansion valve **150** and the chiller **160** are disposed therein.

The compressor **110** operates by receiving power from an engine (internal combustion engine) or a motor. The compressor **110** sucks the refrigerant, compresses the refrigerant into a high-temperature, high-pressure gaseous refrigerant, and then discharges the refrigerant to the water-cooled condenser **120**.

The refrigerant flowing through the water-cooled condenser **120** exchanges heat with the coolant in the heating line **230** and then flows to the first expansion valve **130**. As described above, the coolant heated by the refrigerant passing through the water-cooled condenser **120** may be supplied to the interior heat exchanger through the coolant circulation line.

The refrigerant branch part **101** may distribute the refrigerant flowing along the refrigerant circulation line **100** to the first refrigerant line and the second refrigerant line.

The first expansion valve **130** and the water-cooled evaporator **140** may be disposed in the first refrigerant line.

The first expansion valve **130** may throttle the refrigerant introduced from the first refrigerant line, perform a bypass operation on the refrigerant, or block a flow of the refrigerant. The first expansion valve **130** may be disposed at a side adjacent to the inlet of the water-cooled evaporator **140** based on the flow direction of the refrigerant.

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The water-cooled evaporator **140** is disposed in the first refrigerant line. The refrigerant discharged from the first expansion valve **130** is supplied to the water-cooled evaporator **140** and passes through the water-cooled evaporator **140** while exchanging heat with the coolant passing through the cooling line **220**.

The second expansion valve **150** may throttle the refrigerant introduced through the second refrigerant line, perform a bypass operation on the refrigerant, or block the flow of the refrigerant. The second expansion valve **150** may be disposed at a side adjacent to the inlet of the chiller **160** based on the flow direction of the refrigerant.

The low-temperature, low-pressure refrigerant discharged from the second expansion valve **150** is supplied to the chiller **160** and exchanges heat with the coolant flowing in the coolant line **200**. The cold coolant made by heat exchange in the chiller **160** may flow along the coolant line **200** and exchange heat with the high-temperature battery **213**. That is, the battery **213** exchanges heat with the coolant instead of exchanging heat with the refrigerant.

The accumulator **170** is installed at a side adjacent to the inlet of the compressor **110**. The refrigerant having passed through the evaporator and/or the chiller **160** merges into the accumulator **170**. The accumulator **170** may divide the refrigerant into a liquid refrigerant and a gaseous refrigerant and supply only the gaseous refrigerant to the compressor **110**.

In addition, the refrigerant heat exchanger **180** may be disposed in the refrigerant circulation line **100**.

The refrigerant heat exchanger **180** allows the refrigerant passing through the water-cooled condenser **120** and flowing to the refrigerant branch part **101** and the refrigerant discharged from the accumulator **170** to exchange heat with each other, thereby improving cooling performance. In this case, the inlet side refrigerant line configured to connect the water-cooled condenser **120** and the refrigerant branch part **101** passes through the refrigerant heat exchanger **180**, and the outlet side refrigerant line passes through the accumulator **170** and the compressor **110**, such that the refrigerant passing through the inlet side refrigerant line and the refrigerant passing through the outlet side refrigerant line may exchange heat with each other.

Therefore, the refrigerant may be further cooled by the refrigerant heat exchanger **180** before being divided by the refrigerant branch part, thereby improving cooling performance implemented by the evaporator and improving efficiency of the cooling system.

The refrigerating line **210** may include the radiator **211**, the fourth direction switching valve **212**, the battery **213**, the first pump **214**, and the fifth direction switching valve **215**.

The radiator **211** cools the coolant having exchanged heat with the electrical component **253** and/or the battery **213**. The radiator **211** may be cooled by the cooling fan **211a** in an air-cooled manner.

The fourth direction switching valve **212** may connect the refrigerating line **210** and the cooling line **220** and control the flow of the coolant so that the refrigerating line **210** and the cooling line **220** are connected or disconnected according to the air conditioning mode.

A 4-way valve may be used as the fourth direction switching valve **212**. The two branch points of the fourth direction switching valve **212** may be connected to the refrigerating line **210**, and the remaining two branch points of the fourth direction switching valve **212** may be connected to the cooling line **220**.

In one embodiment, a binary valve may be used as the fourth direction switching valve **212**.

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The fourth direction switching valve **212** may control the flow direction of the coolant on the basis of the air conditioning mode to be described below, thereby improving heating efficiency.

The battery **213** serves as a power source for the vehicle. The battery **213** may serve as a driving source for various types of electrical components **253** in the vehicle. In addition, the battery **213** may be connected to a fuel cell and serve to store electricity. Alternatively, the battery **213** may serve to store electricity supplied from the outside. The battery **213** may be cooled or heated by exchanging heat with the flowing coolant.

The first pump **214** is disposed in the refrigerating line **210** and serves to pump the coolant.

The fifth direction switching valve **215** may be disposed in the refrigerating line **210**, and the third connection line **270** passing through the chiller **160** may be connected to one side of the fifth direction switching valve **215**. The fifth direction switching valve **215** may control the flow of the coolant on the basis of the air conditioning mode.

The first direction switching valve **231**, the water-cooled condenser **120**, the second pump **232**, the coolant heater **233**, and the heater core **234** may be disposed in the heating line **230**.

The first direction switching valve **231** may be connected to the first connection line **250**, the second connection line **260**, and the heating line **230**, and the refrigerating line **210** and the heating line **230** may be connected or disconnected by the first direction switching valve **231**.

More specifically, the first direction switching valve **231** may be installed in the heating line **230**. Two coolant line pipes may be connected to the first direction switching valve **231**. A single first connection line **250** branching off from one side of the refrigerating line **210** may be connected to the first direction switching valve **231**. A single second connection line branching off from the other side of the refrigerating line **210** may be connected to the first direction switching valve **231**. That is, four coolant lines **200** may be connected to the first direction switching valve **231** so as to meet together. The fourth direction switching valve **212** may be a 4-way direction switching valve capable of adjusting states in which the four coolant lines **200** are connected to or disconnected from one another.

The coolant passing through the water-cooled condenser **120** may exchange heat with the refrigerant passing through the water-cooled condenser **120**.

The second pump **232** is a mean disposed in the heating line **230** and configured to pump the coolant so that the coolant flows along the heating line **230**.

The coolant heater **233** refers to a device for heating the coolant. The coolant heater **233** is connected and disposed rearward of the second pump **232** and forward of the heater core **234** based on the flow direction of the coolant. Further, the coolant heater **233** may operate when a temperature of the coolant is equal to or lower than a particular temperature. Various components such as an induction heater, a sheath heater, a PTC heater, or a film heater capable of generating heat using electric power may be used as the coolant heater **233**.

The heater core **234** may be disposed in the air conditioning device of the vehicle. The air flowing by the air blower may be heated while passing through the heater core **234**, supplied to the vehicle interior, and used to heat the vehicle interior. Further, the heater core **234** may be connected and disposed rearward of the coolant heater **233** based on the flow direction of the coolant.

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A cabin cooler **222** is disposed in the cooling line **220**, and the vehicle interior may be cooled by using the coolant passing through the water-cooled evaporator **140**. The third pump **221** and the cabin cooler **222** may be disposed in the cooling line **220**, and the cooling line **220** may be connected to the refrigerating line **210** through the cooling connection line **223**.

The third pump **221** is a means disposed in the cooling line **220** and configured to pump the coolant so that the coolant flows along the cooling line **220**.

The cabin cooler **222** serves as an air-cooled evaporator. The coolant cooled by exchanging heat with the refrigerant passing through the water-cooled evaporator **140** passes through the cabin cooler **222**. Further, the cabin cooler **222** is disposed in the air conditioning device. The air flowing by the air blower of the air conditioning device is cooled while passing through the cabin cooler **222**, supplied to the vehicle interior, and used to cool the vehicle interior.

The cooling connection line **223** may connect the cooling line **220** and the refrigerating line **210**. The cooling connection line **223** may allow the coolant to flow according to the air conditioning mode. The cooling connection line **223** may allow the coolant introduced through the refrigerating line **210** to flow into the refrigerating line **210** again via the water-cooled evaporator **140** disposed in the cooling line **220**.

In one embodiment, one side of the cooling connection line **223** may be disposed between the water-cooled evaporator **140** and the cabin cooler **222**. This is to prevent the coolant flowing along the cooling line **220** from being introduced into the cabin cooler **222** when the refrigerating line **210** and the cooling line **220** are connected by the fourth direction switching valve **212**.

The first connection line **250** may branch off from one side of the refrigerating line **210** and be connected to one region of the heating line **230**.

A reservoir tank **251**, the third pump **221**, and the electrical component **253** may be disposed in the first connection line **250**.

The reservoir tank **251** may serve to store the coolant and supplement the coolant in the coolant line.

In one embodiment, the reservoir tank **251** may be disposed at a connection portion between the first connection line **250** and the refrigerating line **210**. Since the reservoir tank **251** is disposed at the connection portion between the first connection line **250** and the refrigerating line **210** as described above, it is possible to improve efficiency in supplementing or storing the coolant regardless of the flow direction of the coolant according to the air conditioning mode.

The fourth pump **252** may be disposed in the first connection line **250** and pump the coolant.

The electrical component **253** may be disposed in the first connection line **250**. The electrical component **253** may be cooled by the coolant. Further, the electrical component **253** may be a drive motor, an inverter, a charger (onboard charger (OBC)), or the like.

The second connection line **260** may branch off from the other side of the refrigerating line **210** and be connected to one region of the heating line **230**. One side of the second connection line **260** is connected to the first direction switching valve **231** disposed in the heating line **230**, and the other side of the second connection line **260** may be connected to one region of the refrigerating line **210**.

The third connection line **270** may branch off from the fifth coolant joint **271** disposed in the refrigerating line **210**. One side of the third connection line **270** may be connected

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to the refrigerating line **210**, and the other side of the third connection line **270** may be connected to the fifth direction switching valve **215**. The coolant flowing through the third connection line **270** may pass through the chiller **160**.

In this case, the chiller **160** is as described above with reference to the refrigerant circulation line **100**.

The fifth direction switching valve **215** may be disposed in the refrigerating line **210**, and two coolant pipes are connected to the fifth direction switching valve **215**. The fifth direction switching valve **215** may be a 3-way direction switching valve to which the other side of the third connection line **270** is connected.

In one embodiment, the third connection line **270** may be connected in parallel with the refrigerating line **210** in which the battery **213** is disposed.

FIG. **8** is a view illustrating an operation in a first air conditioning mode in FIG. **7**.

Referring to FIG. **8**, the first air conditioning mode is a mode in which the vehicle interior is cooled and the battery **213** is cooled by the chiller **160**.

In the first air conditioning mode, the compressor **110** operates in the refrigerant circulation line **100**, and the high-temperature, high-pressure refrigerant is discharged from the compressor **110**. Further, the refrigerant discharged from the compressor **110** is cooled while exchanging heat with the coolant in the water-cooled condenser **120**. The refrigerant having passed through the water-cooled condenser **120** passes through the refrigerant heat exchanger **180** while exchanging heat with the refrigerant passing through the accumulator **170**.

Thereafter, the refrigerant having passed through the refrigerant heat exchanger **180** is divided in the refrigerant branch part **101**. A part of the refrigerant is throttled and expanded while passing through the first expansion valve **130** disposed in the first refrigerant line. Thereafter, the expanded refrigerant exchanges heat with the coolant in the secondary cooling line **220** while passing through the water-cooled evaporator **140**, and the coolant in the secondary cooling line **220** is cooled by the refrigerant.

Further, the refrigerant evaporated in the water-cooled evaporator **140** is divided into a gaseous refrigerant and a liquid refrigerant while passing through the accumulator **170**, passes through the refrigerant heat exchanger **180**, and then flows into the compressor **110** again.

In addition, the remaining part of refrigerant divided in the refrigerant branch part **101** is throttled and expanded while passing through the second expansion valve **150** disposed in the second refrigerant line. Thereafter, the expanded refrigerant is evaporated by exchanging heat with the coolant while passing through the chiller **160**, such that the coolant may be cooled. Further, the refrigerant evaporated in the chiller **160** is divided into a gaseous refrigerant while passing through the accumulator **170** and then flows into the compressor **110** again via the refrigerant heat exchanger **180**.

As described above, the refrigerant having passed through the water-cooled evaporator **140** and the refrigerant having passed through the chiller **160** merge with each other in the accumulator **170** and flow into the compressor **110**. The refrigerant circulates as the above-mentioned process is repeated.

Meanwhile, the coolant in the coolant line **200** is circulated by the operations of the first to fourth pumps **214**, **232**, **221**, and **252**.

In this case, the fourth direction switching valve **212** may operate so that the refrigerating line **210** and the cooling line **220** are separated. The fifth direction switching valve **215**

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blocks the coolant line connected to the radiator **211**, so that the coolant passing through the battery **213** and the chiller **160** defines the closed loop.

The coolant pumped by the fourth pump **252** disposed in the first connection line **250** passes through the electrical component **253**, and the coolant having passed through the electrical component **253** may pass through the first direction switching valve **231**.

In this case, the first direction switching valve **231** connects the first connection line **250** to one region of the heating line **230**, such that a temperature of the coolant having passed through the electrical component **253** is increased as the coolant passes through the water-cooled condenser **120**. The coolant with the increased temperature is pumped by the second pump **232** and flows into the first direction switching valve **231** via the coolant heater **233** and the heater core **234**.

The coolant introduced into the first direction switching valve **231** after passing through the water-cooled condenser **120** flows along the second connection line **260**. The coolant may be cooled by the cooling fan **211a** while passing through the radiator **211** and then introduced into the fourth direction switching valve **212** again. The coolant having passed through the fourth direction switching valve **212** may flow into the reservoir tank **251** and then circulate through the fourth pump **252**.

The refrigerating line **210** and the cooling line **220** are separated by the fourth direction switching valve **212**.

The coolant in the cooling line **220** circulates through the third pump **221**. The coolant having performed the heat exchange while passing through the water-cooled evaporator **140** passes through the cabin cooler. The coolant passing through the cabin cooler **222** exchanges heat with the air blown by the air blower of the air conditioning device **190**, such that the air is cooled. The cooled air is supplied to the vehicle interior and used to cool the vehicle interior.

In addition, the fifth direction switching valve **215** may operate so that the third connection line **270** is connected to one region of the refrigerating line **210**. Therefore, the fifth direction switching valve **215** may prevent the coolant from flowing from the reservoir tank **251** to the fifth coolant joint **271**.

The coolant flows from the chiller **160** and circulates sequentially through the first pump **214**, the fifth direction switching valve **215**, the fifth coolant joint **271**, and the battery **213**. This cycle is repeated. That is, the battery **213** and the chiller **160** defines a separate closed loop, in which the coolant circulates, by the fifth direction switching valve **215**, such that the battery **213** may be separately cooled by the chiller **160**.

FIG. 9 is a view illustrating an operation in a second air conditioning mode in FIG. 7.

Referring to FIG. 9, the second air conditioning mode is a mode in which the vehicle interior is cooled and the battery **213** is cooled by using the radiator **211**.

In the second air conditioning mode, the compressor **110** operates in the refrigerant circulation line **100**, and the high-temperature, high-pressure refrigerant is discharged from the compressor **110**. Further, the refrigerant discharged from the compressor **110** is cooled while exchanging heat with the coolant in the water-cooled condenser **120**. The refrigerant having passed through the water-cooled condenser **120** passes through the refrigerant heat exchanger **180** while exchanging heat with the refrigerant passing through the accumulator **170**.

Thereafter, the refrigerant having passed through the refrigerant heat exchanger **180** is divided in the refrigerant

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branch part **101**. A part of the refrigerant is throttled and expanded while passing through the first expansion valve **130**. Thereafter, the expanded refrigerant exchanges heat with the coolant in the secondary cooling line **220** while passing through the water-cooled evaporator **140**, and the coolant in the secondary cooling line **220** is cooled by the refrigerant.

Further, the refrigerant evaporated in the water-cooled evaporator **140** is divided into a gaseous refrigerant and a liquid refrigerant while passing through the accumulator **170**, passes through the refrigerant heat exchanger **180**, flows into the compressor **110** again, and then circulates.

In this case, in the second air conditioning mode, the chiller **160** is not used, and the second expansion valve **150** is closed, such that the refrigerant does not flow.

Meanwhile, the coolant in the coolant line **200** is circulated by the operations of the first to fourth pumps **214**, **232**, **221**, and **252**.

In this case, the fourth direction switching valve **212** may operate so that the refrigerating line **210** and the cooling line **220** are separated. The fifth direction switching valve **215** may close the third connection line **270** so that the coolant passing through the battery **213** flows into the radiator **211**.

The coolant pumped by the fourth pump **252** disposed in the first connection line **250** passes through the electrical component **253**, and the coolant having passed through the electrical component **253** may pass through the first direction switching valve **231**.

In this case, the first direction switching valve **231** connects the first connection line **250** to one region of the heating line **230**, such that a temperature of the coolant having passed through the electrical component **253** is increased as the coolant passes through the water-cooled condenser **120**. The coolant with the increased temperature is pumped by the second pump **232** and flows into the first direction switching valve **231** via the coolant heater **233** and the heater core **234**.

The coolant introduced into the first direction switching valve **231** after passing through the water-cooled condenser **120** flows along the second connection line **260**. The coolant may be cooled by the cooling fan **211a** while passing through the radiator **211** and then introduced into the fourth direction switching valve **212** again. The coolant having passed through the fourth direction switching valve **212** may flow into the reservoir tank **251** and then circulate through the fourth pump **252**.

The coolant divided in the reservoir tank **251** flows along the refrigerating line **210**, flows to the battery **213** via the fifth coolant joint **271**, and flows to the fifth direction switching valve **215** via the second pump **232**. In this case, the fifth direction switching valve **215** blocks the flow of the coolant to the third connection line **270**, such that the coolant flows into the radiator **211** and then circulates.

Therefore, the battery **213** may be cooled by the coolant circulating through the radiator **211**.

The refrigerating line **210** and the cooling line **220** are separated by the fourth direction switching valve **212**.

The coolant in the cooling line **220** circulates through the third pump **221**. The coolant having performed the heat exchange while passing through the water-cooled evaporator **140** passes through the cabin cooler. The coolant passing through the cabin cooler **222** exchanges heat with the air blown by the air blower of the air conditioning device **190**, such that the air is cooled. The cooled air is supplied to the vehicle interior and used to cool the vehicle interior.

FIG. 10 is a view illustrating an operation in a third air conditioning mode in FIG. 7.

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Referring to FIG. 10, the third air conditioning mode is a mode in which the vehicle interior is heat, heat of the outside air is absorbed, and a temperature of the battery 213 is increased.

Because the operation of the refrigerant circulation line 100 in the third air conditioning mode is identical to the operation in the first air conditioning mode, a description thereof will be omitted.

One region of the refrigerating line 210 of the coolant line and one region of the cooling line 220 may be connected by the fourth direction switching valve 212.

The fourth direction switching valve 212 may connect the refrigerating line 210 adjacent to the radiator 211 and the cooling line 220 adjacent to the water-cooled evaporator 140 and operate to block the flow of the coolant between the refrigerating line 210 adjacent to the reservoir and the cooling line 220 adjacent to the cabin cooler 222.

The coolant having passed through the radiator 211 flows to the cooling line 220 through the fourth direction switching valve 212. The coolant having passed through the evaporator flows along the cooling connection line 223, flows into the radiator 211, and circulates. The circulation of the coolant may absorb heat from the outside air by means of the water-cooled evaporator 140. In this case, the fourth direction switching valve 212 may block the flow of the coolant flowing from the cooling line 220 to the cabin cooler 222.

The coolant circulating through the fourth pump 252 flows into the first direction switching valve 231 via the electrical component 253. In this case, the first direction switching valve 231 connects the first connection line 250 to one region of the heating line 230, such that a temperature of the coolant having passed through the electrical component 253 is increased as the coolant passes through the water-cooled condenser 120. The coolant with the increased temperature is pumped by the second pump 232 and flows into the first direction switching valve 231 via the coolant heater 233 and the heater core 234.

The coolant introduced into the first direction switching valve 231 after passing through the water-cooled condenser 120 flows along the second connection line 260 and flows to the battery 213. In this case, the fifth direction switching valve 215 may block the flow of the coolant to the third connection line 270.

The coolant having the temperature increased in the water-cooled condenser 120 may increase a temperature of the battery 213 while flowing along the refrigerating line 210, flow into the reservoir tank 251, and then circulate through the refrigerating line 210 and the heating line 230 through the fourth pump 252.

FIG. 11 is a view illustrating an operation in a fourth air conditioning mode in FIG. 7.

Referring to FIG. 11, the fourth air conditioning mode is a mode in which the vehicle interior is heat, heat of the outside air is absorbed, and heat generated by the electrical component 253 and the battery 213 is absorbed.

Because the operation of the refrigerant circulation line 100 in the fourth air conditioning mode is identical to the operation in the first air conditioning mode, a description thereof will be omitted.

One region of the refrigerating line 210 of the coolant line and one region of the cooling line 220 may be connected by the fourth direction switching valve 212.

The fourth direction switching valve 212 may connect the refrigerating line 210 adjacent to the radiator 211 and the cooling line 220 adjacent to the water-cooled evaporator 140 and operate to block the flow of the coolant between the

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refrigerating line 210 adjacent to the reservoir and the cooling line 220 adjacent to the cabin cooler 222.

The coolant having passed through the radiator 211 flows to the cooling line 220 through the fourth direction switching valve 212. The coolant having passed through the evaporator flows along the cooling connection line 223, flows into the radiator 211, and circulates. The circulation of the coolant may absorb heat from the outside air by means of the water-cooled evaporator 140. In this case, the fourth direction switching valve 212 may block the flow of the coolant flowing from the cooling line 220 to the cabin cooler 222.

The coolant circulating through the fourth pump 252 flows into the first direction switching valve 231 via the electrical component 253. In this case, the first direction switching valve 231 may control the flow of the coolant so that the heating line 230 independently circulates.

The coolant having passed through the electrical component 253 flows to the battery 213 along the second connection line 260. In this case, the fifth direction switching valve 215 may open the third connection line 270 so that the coolant may flow to the chiller 160.

The coolant flowing along the third connection line 270 may be cooled by heat exchange in the chiller 160, divided in the fifth coolant joint 271, and then moved.

The coolant circulating toward the reservoir tank 251 may absorb heat of the electrical component 253, and the coolant circulating toward the battery 213 may absorb heat of the battery 213.

The coolant in the heating line 230 absorbs heat from the water-cooled condenser 120. The coolant passing through the water-cooled condenser may heat the vehicle interior while defining a closed loop that circulates through the first direction switching valve 231 while passing through the second pump 232, the coolant heater 233, and the heater core 234.

FIG. 12 is a view illustrating an operation in a fifth air conditioning mode in FIG. 7.

Referring to FIG. 12, the fifth air conditioning mode is a mode in which the vehicle interior is heated and dehumidified, and heat is absorbed from the outside air, the electrical component 253, and the battery.

Because the operation of the refrigerant circulation line 100 in the fifth air conditioning mode is identical to the operation in the first air conditioning mode, a description thereof will be omitted.

Meanwhile, the coolant in the coolant line 200 is circulated by the operations of the first to fourth pumps 214, 232, 221, and 252.

The fourth direction switching valve 212 may operate so that the refrigerating line 210 and the cooling line 220 are separated.

The coolant pumped by the fourth pump 252 disposed in the first connection line 250 passes through the electrical component 253, and the coolant having passed through the electrical component 253 may pass through the first direction switching valve 231.

The coolant circulating through the fourth pump 252 flows into the first direction switching valve 231 via the electrical component 253. In this case, the first direction switching valve 231 may control the flow of the coolant so that the heating line 230 independently circulates.

The coolant having passed through the electrical component 253 flows along the second connection line 260 and flows by being divided in the region in which the second connection line 260 meets the refrigerating line 210.

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The coolant flowing from the second connection line 260 toward the battery 213 flows into the fifth direction switching valve 215. The fifth direction switching valve 215 may open the third connection line 270 so that the coolant may flow to the chiller 160.

The coolant flowing along the third connection line 270 may be cooled by heat exchange in the chiller 160, divided in the fifth coolant joint 271, and then moved.

The coolant, which is divided in the fifth coolant joint 271 and circulates toward the reservoir tank 251, may absorb heat of the electrical component 253 while circulating through the first pump 214.

In addition, the coolant, which is divided in the fifth coolant joint 271 and circulates toward the battery 213, may absorb heat of the battery 213 and then flow to the fifth direction switching valve 215.

The coolant, which is divided in the second connection line 260 and circulates toward the radiator 211, is cooled by the cooling fan 211a while passing through the radiator 211, passes through the fourth direction switching valve 212, and then flows into the reservoir tank 251.

The coolant in the cooling line 220 circulates through the third pump 221, and the coolant passes through the cabin cooler. The coolant passing through the cabin cooler 222 exchanges heat with the air blown by the air blower of the air conditioning device, and moisture in the air is removed as the coolant and the air exchange heat with each other.

Meanwhile, the coolant in the heating line 230 absorbs heat from the water-cooled condenser 120. The coolant passing through the water-cooled condenser 120 may heat the vehicle interior while defining a closed loop that circulates through the first direction switching valve 231 while passing through the second pump 232, the coolant heater 233, and the heater core 234.

FIG. 13 is a view illustrating an operation in a sixth air conditioning mode in FIG. 7.

Referring to FIG. 13, the sixth air conditioning mode is a mode in which the vehicle interior is heated by using waste heat.

In the sixth air conditioning mode, the operation of the refrigerant circulation line 100 is stopped.

One region of the refrigerating line 210 of the coolant line 200 and one region of the cooling line 220 may be connected by the fourth direction switching valve 212. The fourth direction switching valve 212 may connect the refrigerating line 210 adjacent to the radiator 211 and the cooling line 220 adjacent to the water-cooled evaporator 140 and operate to block the flow of the coolant between the refrigerating line 210 adjacent to the reservoir and the cooling line 220 adjacent to the cabin cooler 222. However, since the third pump 221 does not operate, the operation of the coolant line 200 connected through the fourth direction switching valve 212 may be stopped.

The coolant circulating through the fourth pump 252 flows into the first direction switching valve 231 via the electrical component 253. In this case, the first direction switching valve 231 connects the first connection line 250 and one region of the heating line 230 so that the coolant having passed through the electrical component 253 passes through the water-cooled condenser 120.

However, since the refrigerant circulation line 100 does not operate, the coolant passes through the water-cooled condenser 120, and the coolant is pumped by the second pump 232 and flows into the first direction switching valve 231 via the coolant heater 233 and the heater core 234.

The coolant introduced into the first direction switching valve 231 after passing through the water-cooled condenser

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120 flows along the second connection line 260 and flows to the battery 213. In this case, the fifth direction switching valve may block the flow of the coolant to the third connection line 270.

The coolant having passed through the fifth direction switching valve 215 flows along the refrigerating line 210, and absorbs heat of the battery 213. The coolant may flow into the reservoir tank 251 and then circulate through the refrigerating line 210 and the heating line 230 through the fourth pump 252.

In this case, since the refrigerant circulation line 100 is stopped, the coolant heats the vehicle interior by using waste heat of the electrical component 253 and the battery 213.

FIGS. 14 to 20 relates to a vehicle heat pump system according to a third embodiment of the present invention.

FIG. 14 is a structural view of a vehicle heat pump system according to a third embodiment of the present invention. Referring to FIG. 14, the vehicle heat pump system according to the present invention differs from the vehicle heat pump systems according to the above-mentioned embodiments in terms of arrangement structures of the battery 213 and the pump.

Because the refrigerant circulation line 100 of the vehicle heat pump system according to yet another embodiment of the present invention is identical in structure to the above-mentioned refrigerant circulation line 100, a description thereof will be omitted.

Because the coolant line is also identical in structure to the above-mentioned coolant line, only the difference will be described.

The fourth pump 252 disposed in the first connection line 250 may be eliminated according to the operation of the system.

The battery 213 and the first pump 214, which are disposed in the refrigerating line 210 in the above-mentioned embodiment, may be disposed in the third connection line 270.

The battery 213, the chiller 160, and the first pump 214 may be disposed in the third connection line 270, and the fourth connection line 280 may be disposed between the battery 213 and the chiller 160.

One side of the fourth connection line 280 may be disposed in the second connection line 260, and the other side of the fourth connection line 280 may be disposed in the third connection line 270.

The third direction switching valve 261 may be disposed in the region in which the fourth connection line 280 and the second connection line 260 are connected. The second connection line 260 may be connected to two opposite sides of the third direction switching valve 261, and the fourth connection line 280 may be disposed in one region of the third direction switching valve 261.

In addition, the battery 213 may be disposed at the upper side of the third connection line 270, and the first pump 214 and the chiller 160 may be disposed at the lower side of the third connection line 270. The operation of the system according to the air conditioning mode will be described on the basis of the structural difference illustrated in FIG. 14.

The first to sixth air conditioning modes are identical to the above-mentioned modes. Because the refrigerant circulation line 100 according to the air conditioning mode of the vehicle heat pump system according to yet another embodiment of the present invention to be described below is identical to the above-mentioned refrigerant circulation line, the following description will be made on the basis of the coolant line.

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FIG. 15 is a view illustrating an operation in a first air conditioning mode in FIG. 14.

Referring to FIG. 15, in the first air conditioning mode, the coolant line 200 identical to that of the above-mentioned embodiment operates. In this case, the third direction switching valve 261 closes the fourth connection line 280, so that the operation identical to the operation of the coolant line 200 of the vehicle heat pump system according to the above-mentioned embodiment is performed.

FIG. 16 is a view illustrating an operation in a second air conditioning mode in FIG. 14.

Referring to FIG. 16, the coolant in the coolant line 200 is circulated by the operations of the first to fourth pumps 214, 232, 221, and 221.

In this case, the fourth direction switching valve 212 may operate so that the refrigerating line 210 and the cooling line 220 are separated. The fifth direction switching valve 215 may operate to connect the third connection line 270 and close one region of the refrigerating line 210.

The coolant having passed through the electrical component 253 disposed in the first connection line 250 may pass through the first direction switching valve 231.

In this case, the first direction switching valve 231 connects the first connection line 250 to one region of the heating line 230, such that a temperature of the coolant having passed through the electrical component 253 is increased as the coolant passes through the water-cooled condenser 120. The coolant with the increased temperature is pumped by the second pump 232 and flows into the first direction switching valve 231 via the coolant heater 233 and the heater core 234.

The coolant introduced into the first direction switching valve 231 after passing through the water-cooled condenser 120 flows along the second connection line 260. The coolant may be cooled by the cooling fan 211a while passing through the radiator 211 and then introduced into the fourth direction switching valve 212 again. The coolant having passed through the fourth direction switching valve 212 may flow into the reservoir tank 251 and then circulate.

The coolant divided in the reservoir tank 251 flows along the refrigerating line 210. In this case, the coolant may flow along the third connection line 270 through the fifth direction switching valve 215 and the fifth coolant joint 271, flow through the battery 213, the chiller 160, and the first pump 214 disposed in the third connection line 270, and then flow to the radiator 211 along the refrigerating line 210.

In this case, the second refrigerant line of the refrigerant circulation line 100 does not operate. Therefore, the chiller 160 does not operate, the coolant passes through the chiller 160, and the battery 213 is cooled by the radiator 211.

Meanwhile, the coolant in the cooling line 220 circulates through the third pump 221. The coolant having performed the heat exchange while passing through the water-cooled evaporator 140 passes through the cabin cooler. The coolant passing through the cabin cooler 222 exchanges heat with the air blown by the air blower of the air conditioning device 190, such that the air is cooled. The cooled air is supplied to the vehicle interior and used to cool the vehicle interior.

FIG. 17 is a view illustrating an operation in a third air conditioning mode in FIG. 14.

Referring to FIG. 17, in the third air conditioning mode, one region of the refrigerating line 210 of the coolant line 200 and one region of the cooling line 220 may be connected by the fourth direction switching valve 212.

The fourth direction switching valve 212 may connect the refrigerating line 210 adjacent to the radiator 211 and the cooling line 220 adjacent to the water-cooled evaporator 140

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and operate to block the flow of the coolant between the refrigerating line 210 adjacent to the reservoir and the cooling line 220 adjacent to the cabin cooler 222.

The coolant having passed through the radiator 211 flows to the cooling line 220 through the fourth direction switching valve 212. The coolant having passed through the evaporator flows along the cooling connection line 223, flows into the radiator 211, and circulates. The circulation of the coolant may absorb heat from the outside air by means of the water-cooled evaporator 140. In this case, the fourth direction switching valve 212 may block the flow of the coolant flowing from the cooling line 220 to the cabin cooler 222.

The coolant flows into the first direction switching valve 231 via the electrical component 253. In this case, the first direction switching valve 231 connects the first connection line 250 to one region of the heating line 230, such that a temperature of the coolant having passed through the electrical component 253 is increased as the coolant passes through the water-cooled condenser 120. The coolant with the increased temperature is pumped by the second pump 232 and flows into the first direction switching valve 231 via the coolant heater 233 and the heater core 234.

The coolant introduced into the first direction switching valve 231

after passing through the water-cooled condenser 120 flows along the second connection line 260.

In this case, the third direction switching valve 261 operates so that the fourth connection line 280 and the second connection line 260 are connected, such that the coolant flows along the fourth line. The fifth direction switching valve 215 blocks the flow in the refrigerating line 210, such that the coolant flows to the battery 213.

The coolant having the temperature increased in the water-cooled condenser 120 may flow to the battery 213 in the third connection line 270 through the third direction switching valve 261, increase the temperature of the battery 213, flow into the reservoir tank 251, and then circulate through the refrigerating line 210 and the heating line 230.

In the structure described above, the coolant may bypass the first pump 214 when the temperature of the battery 213 increases, thereby preventing a reverse rotation of the first pump 214.

FIG. 18 is a view illustrating an operation in a fourth air conditioning mode in FIG. 14.

Referring to FIG. 18, in the fourth air conditioning mode, one region of the refrigerating line 210 of the coolant line 200 and one region of the cooling line 220 may be connected by the fourth direction switching valve 212.

The fourth direction switching valve 212 may connect the refrigerating line 210 adjacent to the radiator 211 and the cooling line 220 adjacent to the water-cooled evaporator 140 and operate to block the flow of the coolant between the refrigerating line 210 adjacent to the reservoir and the cooling line 220 adjacent to the cabin cooler 222.

The coolant having passed through the radiator 211 flows to the cooling line 220 through the fourth direction switching valve 212. The coolant having passed through the evaporator flows along the cooling connection line 223, flows into the radiator 211, and circulates. The circulation of the coolant may absorb heat from the outside air by means of the water-cooled evaporator 140. In this case, the fourth direction switching valve 212 may block the flow of the coolant flowing from the cooling line 220 to the cabin cooler 222.

The coolant having passed through the electrical component 253 disposed in the first connection line 250 flows into

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the first direction switching valve 231. In this case, the first direction switching valve 231 connects the first connection line 250 to one region of the heating line 230, such that a temperature of the coolant having passed through the electrical component 253 is increased as the coolant passes through the water-cooled condenser 120. The coolant with the increased temperature is pumped by the second pump 232 and flows into the first direction switching valve 231 via the coolant heater 233 and the heater core 234.

The coolant introduced into the first direction switching valve 231 after passing through the water-cooled condenser 120 flows along the second connection line 260.

In this case, the third direction switching valve 261 operates so that the fourth connection line 280 and the second connection line 260 are connected, such that the coolant flows along the fourth connection line 280 and is divided in the third connection line 270.

The coolant divided in the third connection line 270 to the battery 213 passes through the battery 213, flows into the reservoir tank 251, and then circulates.

The coolant divided in the third connection line 270 to the chiller 160 performs heat exchange in the chiller 160, flows along the refrigerating line 210 via the first pump 214 and the fifth direction switching valve 215, and flows into the reservoir tank 251.

The coolant flowing along the coolant line 200 may absorb waste heat of the electrical component 253 and the battery 213.

FIG. 19 is a view illustrating an operation in a fifth air conditioning mode in FIG. 14.

Referring to FIG. 19, in the fifth air conditioning mode, the coolant in the coolant line 200 is circulated by the operations of the first to fourth pumps 214, 232, 221, and 221.

The fourth direction switching valve 212 may operate so that the refrigerating line 210 and the cooling line 220 are separated.

The coolant may pass through the electrical component 253 disposed in the first connection line 250. The coolant having passed through the electrical component 253 may pass through the first direction switching valve 231.

In this case, the first direction switching valve 231 may control the flow of the coolant so that the heating line 230 independently circulates.

The coolant passing through the electrical component 253 may flow along the second connection line 260 and pass through the third direction switching valve 261.

In this case, the third direction switching valve 261 operates so that the fourth connection line 280 and the second connection line 260 are connected, such that the coolant flows along the fourth connection line 280 and is divided in the third connection line 270.

The coolant divided in the third connection line 270 to the battery 213 passes through the battery 213, flows into the reservoir tank 251, and then circulates.

The coolant divided in the third connection line 270 to the chiller 160 performs heat exchange in the chiller 160, flows along the refrigerating line 210 via the first pump 214 and the fifth direction switching valve 215, and flows to the radiator 211. The coolant flowing to the radiator 211 may be cooled by the cooling fan 211a and flow into the reservoir tank 251 via the fourth direction switching valve 212.

The coolant in the cooling line 220 circulates through the third pump 221, and the coolant passes through the cabin cooler. The coolant passing through the cabin cooler 222 exchanges heat with the air blown by the air blower of the

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air conditioning device, and moisture in the air is removed as the coolant and the air exchange heat with each other.

Meanwhile, the coolant in the heating line 230 absorbs heat from the water-cooled condenser 120. The coolant passing through the water-cooled condenser 120 may heat the vehicle interior while defining a closed loop that circulates through the first direction switching valve 231 while passing through the second pump 232, the coolant heater 233, and the heater core 234.

FIG. 20 is a view illustrating an operation in a sixth air conditioning mode in FIG. 14.

Referring to FIG. 20, in the sixth air conditioning mode, the operation of the refrigerant circulation line 100 is stopped.

One region of the refrigerating line 210 of the coolant line 200 and one region of the cooling line 220 may be connected by the fourth direction switching valve 212. The fourth direction switching valve 212 may connect the refrigerating line 210 adjacent to the radiator 211 and the cooling line 220 adjacent to the water-cooled evaporator 140 and operate to block the flow of the coolant between the refrigerating line 210 adjacent to the reservoir and the cooling line 220 adjacent to the cabin cooler 222. However, since the third pump 221 does not operate, the operation of the coolant line 200 connected through the fourth direction switching valve 212 may be stopped.

The coolant having passed through the electrical component 253 disposed in the first connection line 250 flows into the first direction switching valve 231. In this case, the first direction switching valve 231 connects the first connection line 250 and one region of the heating line 230 so that the coolant having passed through the electrical component 253 passes through the water-cooled condenser 120.

However, since the refrigerant circulation line 100 does not operate, the coolant passes through the water-cooled condenser 120, and the coolant is pumped by the second pump 232 and flows into the first direction switching valve 231 via the coolant heater 233 and the heater core 234.

The coolant introduced into the first direction switching valve 231 after passing through the water-cooled condenser 120 flows along the second connection line 260.

In this case, the third direction switching valve 261 operates so that the fourth connection line 280 and the second connection line 260 are connected, such that the coolant flows along the fourth connection line 280. In this case, the fifth direction switching valve 215 may block the flow of the coolant in the refrigerating line 210. Therefore, the coolant flowing along the third connection line 270 passes through the battery 213 and then flows from the fifth coolant joint 271 to the reservoir tank 251.

In this case, since the refrigerant circulation line 100 is stopped, the coolant heats the vehicle interior by using waste heat of the electrical component 253 and the battery 213.

The components indicated by the same reference numerals in the description of the vehicle heat pump system according to the embodiments of the present invention may perform the same functions, and the configurations and contents related to the embodiments may be carried out interchangeably.

FIGS. 21 to 25 relates to a vehicle heat pump system according to a fourth embodiment of the present invention.

FIG. 21 is a structural view of a vehicle heat pump system according to a fourth embodiment of the present invention.

Referring to FIG. 21, the vehicle heat pump system according to the fourth embodiment of the present invention may include: the compressor 110 configured to compress and circulate the refrigerant; a first heat exchanger 120

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configured to condense the compressed refrigerant; a first expansion valve **130** configured to expand the condensed refrigerant; a second heat exchanger **140** configured to evaporate the refrigerant expanded by the first expansion valve **130**; a second expansion valve **150** configured to expand the condensed refrigerant; the chiller **160** configured to allow the coolant to exchange heat with the refrigerant expanded by the second expansion valve **150**; a heat-dissipation heat exchanger **211** configured to allow the coolant to exchange heat with outside air; and the electrical component **253** configured to be cooled by the coolant. In the heating mode, the coolant having passed through the heat-dissipation heat exchanger **211** may exchange heat with the electrical component **253** and then flow into the chiller **160**, thereby improving heating performance.

Hereinafter, the condenser **120** will be described as an example of the first heat exchanger, the evaporator **140** will be described as an example of the second heat exchanger, the radiator **211** will be described as an example of the heat-dissipation heat exchanger, and the battery **213** will be described as an example of the heat generating component.

The vehicle heat pump system according to the embodiment of the present invention may include the refrigerant circulation line **100** configured to circulate the refrigerant and cool the vehicle interior; and the coolant circulation line **200** configured to circulate the coolant, heat the vehicle interior, and cool the components. Further, the coolant circulation line **200** may include the heating line **230** configured to heat the vehicle interior, and the refrigerating line **210** configured to cool the electrical component **253** and the battery **213**.

The refrigerant circulation line **100** may include the compressor **110**, the condenser **120**, the refrigerant branch part **101**, the first refrigerant line **100a** branching off from the refrigerant branch part **101** toward one side and configured to pass through the first expansion valve and the evaporator **140**, the second refrigerant line **100b** branching off from the refrigerant branch part **101** toward the other side and configured to pass through the second expansion valve **160** and the chiller **160**, and the accumulator **170** through which the refrigerant having passed through the first refrigerant line **100a** and the second refrigerant line **100b** passes. The refrigerant having passed through the accumulator **170** flows into the compressor **110** again, such that refrigerant circulation is implemented.

In this case, the refrigerant heat exchanger **180** may be disposed in the first refrigerant line **100a**. The refrigerant heat exchanger may be disposed between the refrigerant branch part **101** and the first expansion valve **130**, such that the refrigerant introduced into the first expansion valve **130** may exchange heat with the refrigerant having passed through the evaporator **140**.

The compressor **110** operates by receiving power from an engine (internal combustion engine) or a motor. The compressor sucks the refrigerant, compresses the refrigerant into a high-temperature, high-pressure gaseous refrigerant, and then discharges the refrigerant to the condenser **120**.

The condenser **120** serves as a condenser in both the cooling mode and the heating mode. The refrigerant flowing through the condenser **120** exchanges heat with the coolant in the coolant circulation line **200** to be described below and then flows to the first expansion valve **130**. As described above, the coolant heated by the refrigerant in the condenser **120** may be supplied to an interior heat exchanger through the coolant circulation line **200**. In one embodiment, a water-cooled condenser **120** may be used as the condenser **120**.

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The refrigerant branch part **101** distributes the refrigerant to the first refrigerant line **100a** and the second refrigerant line **100b**.

The first expansion valve **130** and the evaporator **140** may be disposed in the first refrigerant line **100a**.

The first expansion valve **130** may throttle the refrigerant introduced from the first refrigerant line **100a**, perform a bypass operation on the refrigerant, or block a flow of the refrigerant. The first expansion valve **130** may be disposed at a side adjacent to the inlet of the evaporator **140** based on the flow direction of the refrigerant.

The evaporator **140** is installed in an air conditioning casing and disposed in the first refrigerant line **100a**. During a process in which the refrigerant discharged from the first expansion valve **130** is supplied to the evaporator **140** and air flowing in the air conditioning casing by a blower passes through the evaporator **140**, the air exchanges heat with the low-temperature, low-pressure refrigerant in the evaporator **140** and is converted into cold air. The cold air is discharged into a vehicle interior and cools an occupant compartment.

The second expansion valve **160** and the chiller **160** may be disposed in the second refrigerant line **100b**.

The second expansion valve **160** may throttle the refrigerant introduced from the second refrigerant line **100b**, perform a bypass operation on the refrigerant, or block a flow of the refrigerant. The second expansion valve may be disposed at a side adjacent to the inlet of the chiller **160** based on the flow direction of the refrigerant.

The low-temperature, low-pressure refrigerant discharged from the second expansion valve **160** is supplied to the chiller **160** and exchanges heat with the coolant flowing in the coolant circulation line **200**. The cold coolant made by heat exchange in the chiller **160** may circulate through the coolant circulation line **200** and exchange heat with the high-temperature battery **213**. That is, the battery **213** exchanges heat with the coolant instead of exchanging heat with the refrigerant.

The accumulator **170** is installed at a side adjacent to an inlet of the compressor **110**. The refrigerant having passed through the evaporator **140** and/or the chiller **160** merges into the accumulator **170**. The accumulator **170** may divide the refrigerant into a liquid refrigerant and a gaseous refrigerant and supply only the gaseous refrigerant to the compressor **110**.

The refrigerant heat exchanger **180** allows the refrigerant introduced into the first expansion valve **130** and the refrigerant discharged from the evaporator **140** to exchange heat with each other, thereby improving cooling performance. In this case, an inlet side refrigerant line configured to connect the refrigerant branch part **101** and the first expansion valve **130** and configured such that the refrigerant is introduced into the evaporator **140** therethrough passes through the refrigerant heat exchanger **180**, and an outlet side refrigerant line configured to connect the evaporator **140** and the accumulator **170** and configured such that the refrigerant is discharged from the evaporator **140** therethrough passes through the refrigerant heat exchanger **180**, such that the refrigerant passing through the inlet side refrigerant line and the refrigerant passing through the outlet side refrigerant line may exchange heat with each other.

Therefore, before the refrigerant is introduced into the first expansion valve **130**, the refrigerant may be further cooled by the refrigerant heat exchanger **180**, thereby improving cooling performance implemented by the evaporator **140** and improving efficiency of the cooling system. In particular, the refrigerant heat exchanger **180** and the chiller **160** are connected in parallel.

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That is, the refrigerant heat exchanger **180** is not disposed in series between the condenser **120** and the chiller **160** in the refrigerant line, but the refrigerant heat exchanger **180** is disposed adjacent to the evaporator **140**, such that the refrigerant heat exchanger **180** and the evaporator **140** may be disposed and connected in series. If the refrigerant heat exchanger is disposed in series between the water-cooled condenser **120** and the chiller **160**, the heating performance may decrease because a pressure drop occurs at a low-pressure side in the heating mode. On the contrary, when the refrigerant heat exchanger is connected in parallel, both the cooling performance and the heating performance increase. This is because there is no refrigerant heat exchanger between the condenser **120** and the chiller **160** in the flow of the refrigerant in the heating mode.

The coolant circulation line **200** may include the heating line **230** configured to heat the vehicle interior, and the refrigerating line **210** and the second cooling line configured to cool the electrical component **253** and the battery **213**.

In this case, the cooling line **220** may include the evaporator **140**, the cabin cooler **222**, and the third pump **221**. In this case, the coolant line, which connects the evaporator **140**, the cabin cooler **222**, and the third pump **221** may define a closed loop.

As described above, the refrigerant and the coolant may exchange heat with each other while passing through the evaporator **140**.

The cabin cooler **222** serves as an air-cooled evaporator **140**. The coolant cooled by exchanging heat with the refrigerant passing through the evaporator **140** passes through the cabin cooler **222**. Further, the cabin cooler **222** is disposed in the air conditioning device **190**. The air flowing by the air blower of the air conditioning device **190** is cooled while passing through the cabin cooler **222**, supplied to the vehicle interior, and used to cool the vehicle interior.

The third pump **221** serves to pump the coolant so that the coolant circulates along the cooling line **220**. Further, the third pump **221** may be installed in the coolant line and disposed between the evaporator **140** and the cabin cooler **222**. The operation of the third pump **221** may circulate the coolant.

Therefore, the vehicle heat pump system according to the present invention adopts the cooling line **220** using the coolant for cooling the vehicle interior, such that the refrigerant circulation line **100** through which the refrigerant circulates may be disposed outside the vehicle interior instead of being disposed in the vehicle interior. Therefore, it is possible to reduce a length of the refrigerant pipe, reduce the amount of refrigerant, and modularize the components in the refrigerant circulation line **100**. In addition, a high-efficiency natural refrigerant may be used as the refrigerant used for the refrigerant circulation line **100**, which may improve the efficiency of the thermal management system.

The heating line **230** may include the condenser **120**, a second pump **232**, a coolant heater **233**, a heater core **234**, and a first direction switching valve **231**.

As described above, the refrigerant and the coolant may exchange heat with each other while passing through the condenser **120**.

The second pump **232** is a means for pumping the coolant so that the coolant circulates along the heating line **230**. The second pump **232** may be installed in the coolant line and disposed rearward of the condenser **120** based on the flow direction of the coolant.

The coolant heater **233** refers to a device for heating the coolant. The coolant heater **233** is connected and disposed rearward of the second pump **232** and forward of the heater

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core **234** based on the flow direction of the coolant. Further, the coolant heater **233** may operate when a temperature of the coolant is equal to or lower than a particular temperature. Various components such as an induction heater, a sheath heater, a PTC heater, or a film heater capable of generating heat using electric power may be used as the coolant heater **233**.

The heater core **234** may be disposed in an air conditioning device **190** of the vehicle. The air flowing by the air blower may be heated while passing through the heater core **234**, supplied to the vehicle interior, and used to heat the vehicle interior. Further, the heater core **234** may be connected and disposed rearward of the coolant heater **233** based on the flow direction of the coolant.

The first direction switching valve **231** may be installed between the heater core **234** and the condenser **120** and configured to selectively connect or disconnect the heating line **230** and the refrigerating line **210** to be described below.

More specifically, the first direction switching valve **231** may be installed in the heating line **230**. Two coolant line pipes may be connected to the first direction switching valve **231**. The single first connection line **250** branching off from one side of the refrigerating line **210** may be connected to the first direction switching valve **231**. The single second connection line **260** branching off from the other side of the refrigerating line **210** may be connected to the first direction switching valve **231**. That is, four coolant lines may be connected to the first direction switching valve **231** so as to meet together. The first direction switching valve **231** may be a 4-way direction switching valve capable of adjusting states in which the four coolant lines are connected to or disconnected from one another.

The refrigerating line **210** may include a radiator **211**, a fourth direction switching valve **212**, a fourth pump **252**, a first direction switching valve **231**, the electrical component **253**, a fifth coolant joint **271**, a sixth coolant joint **262**, a first pump **214**, the battery **213**, the chiller **160**, and a fifth direction switching valve **215**.

The radiator **211** cools the coolant having exchanged heat with the electrical component **253** or the battery **213**. The radiator **211** may be cooled by a cooling fan **211a** in an air-cooled manner.

The fourth direction switching valve **212** may be installed in the refrigerating line **210**. Two coolant pipes may be connected to the fourth direction switching valve **212**. The first direction switching valve **231** and the fourth direction switching valve **212** may be connected by the first connection line **250** so that the heating line **230** and the refrigerating line **210** are connected.

That is, three coolant lines may be connected to the fourth direction switching valve **212** so as to meet together. The fourth direction switching valve **212** may be a 3-way direction switching valve capable of adjusting states in which the three coolant lines are connected to or disconnected from one another.

The fourth pump **252** is a means for pumping the coolant so that the coolant circulates along the refrigerating line **210**. Further, the fourth pump **252** is installed in the first connection line **250** and disposed between the first direction switching valve **231** and the fourth direction switching valve **212**. The operation of the fourth pump **252** may allow the coolant to flow from the fourth direction switching valve **212** to the first direction switching valve **231**.

The first direction switching valve **231** is as described above with reference to the heating line **230**.

The electrical component **253** is disposed in a second connection line **260** that connects the first direction switch-

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ing valve **231** and the sixth coolant joint **262**. The electrical component **253** may be cooled by the coolant. Further, the electrical component **253** may be a drive motor, an inverter, a charger (onboard charger (OBC)), or the like.

The first pump **214** is a means for pumping the coolant so that the coolant circulates along the refrigerating line **210**. Further, the first pump **214** is installed in the coolant line and disposed between the fifth coolant joint **271** and the battery **213**, such that the coolant may flow from the first pump **214** to the battery **213**.

The battery **213** serves as a power source for the vehicle. The battery **213** may serve as a driving source for various types of electrical components **253** in the vehicle. In addition, the battery **213** may be connected to a fuel cell and serve to store electricity. Alternatively, the battery **213** may serve to store electricity supplied from the outside. Further, the battery **213** may be disposed in the coolant line and provided between the first pump **214** and the fifth direction switching valve **215**. Therefore, the battery **213** may be cooled or heated by exchanging heat with the flowing coolant.

The fifth coolant joint **271** is installed in the coolant line and disposed rearward of the fourth direction switching valve **212** based on the flow direction of the coolant. Three coolant lines are connected to the fifth coolant joint **271** so as to meet together. That is, the fifth coolant joint **271** may be installed such that two opposite sides thereof are connected to the refrigerating line **210**, and a third connection line **270** may be connected to a lower side of the fifth coolant joint **271**. In this case, the third connection line **270** may be connected to pass through the chiller **160**.

The sixth coolant joint **262** may be installed at a point at which a rear end of the second connection line **260** meets the refrigerating line **210**. Three coolant lines are connected to the sixth coolant joint **262** so as to meet together. That is, the sixth coolant joint **262** may be installed such that two opposite sides thereof are connected to the refrigerating line **210**, and the second connection line **260** may be connected to an upper side of the sixth coolant joint **262**.

The chiller **160** is as described above with reference to the heating line **230**.

The fifth direction switching valve **215** may be installed in the coolant line and disposed between the battery **213** and the sixth coolant joint **262**. Two coolant pipes may be connected to the fifth direction switching valve **215**. The third connection line **270** may be connected to an upper side of the fifth direction switching valve **215**, such that the battery **213** and the third connection line **270** may be connected in parallel. In this case, the fourth direction switching valve **212** may be a 3-way direction switching valve capable of adjusting states in which the three coolant lines are connected to or disconnected from one another.

Further, the air blower may be installed at one side of the air conditioning device **190** to blow air. A temperature adjustment door may be installed in the air conditioning device **190**. In addition, the evaporator **140** and the heater core **234** disposed in the air conditioning device **190** may be disposed and configured such that on the basis of the operation of the temperature adjustment door, the air discharged from the air blower may flow into the vehicle interior while passing only through the evaporator **140** or flow into the vehicle interior while passing through the evaporator **140** and then passing through the heater core **234**.

In addition, the third direction switching valve **261** may be disposed in the second connection line **260** and control the flow direction of the coolant having passed through the electrical component **253**. The third direction switching

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valve **261** is disposed in the fourth connection line **280** branching off from the second connection line **260**. The fourth connection line **280** may be connected to a fourth coolant joint **272** disposed in the third connection line **270**.

The fourth coolant joint **272** is disposed at a side adjacent to an inlet of the chiller **160** through which the coolant is introduced. The coolant having passed through the electrical component **253** may be introduced into the chiller **160** according to the operation of the third direction switching valve **261**.

In one embodiment, when the upper side and the right side of the third direction switching valve **261** are connected, the electrical component **253** and the chiller **160** may be connected in series.

FIG. **22** is a view illustrating a first embodiment of FIG. **21**.

Referring to FIG. **22**, the vehicle heat pump system illustrated in FIG. **22** has a structure in which the fifth coolant joint **271** and the fourth coolant joint **272** are connected as a single component.

An integrated coolant joint **273** configured by connecting the fifth coolant joint **271** and the fourth coolant joint **272** may be connected to four coolant flow paths.

The integrated coolant joint **273** may be connected to the chiller **160**, the first pump **214**, the fourth direction switching valve **212**, and the fourth connection line **280** connected to the third direction switching valve **261**. The flow paths of the integrated coolant joint **273** may be controlled by the fourth direction switching valve **212**, the fifth direction switching valve **215**, and the third direction switching valve **261**.

FIG. **23** is a view illustrating an operating state of the system in a maximum heating mode.

Referring to FIG. **23**, in the refrigerant circulation line **100**, the compressor **110** operates, and the high-temperature, high-pressure refrigerant is discharged from the compressor **110**. Further, the refrigerant discharged from the compressor **110** is cooled while exchanging heat with the coolant in the condenser **120**. Thereafter, the refrigerant cooled in the condenser **120** passes through the refrigerant branch part **101** and is throttled and expanded while passing through the second expansion valve **160**. Thereafter, the expanded refrigerant flows into the chiller **160** and exchanges heat with the coolant in the chiller **160**, such that the coolant may be cooled, and the refrigerant may be heated. Next, the refrigerant having passed through the chiller **160** flows into the compressor **110** again via the accumulator **170**.

In this case, the first expansion valve **130** is closed, such that the refrigerant may not flow to the evaporator **140**. Therefore, the refrigerant circulates as the above-mentioned process is repeated.

In addition, in the cooling line **220**, the third pump **221** does not operate, such that the coolant may not flow in the cooling line **220**.

Meanwhile, the coolant in the coolant circulation line **200** is circulated by the operations of the second pump **232** and the fourth pump **252**. Further, the coolant may be heated while passing through the condenser **120**, heated by the coolant heater **233**, and heated by waste heat of the electrical component **253**. The coolant may be cooled while passing through the chiller **160**. In this case, the first direction switching valve **231** and the fourth direction switching valve **212** may adjust the directions thereof so that the heating line **230** and the refrigerating line **210** are separated. More specifically, the coolant may flow as the upper side and the right side of the first direction switching valve **231** are connected to each other, and the coolant may flow as the

lower side and the left side of the first direction switching valve **231** are connected to each other. Further, when the left side and the lower side of the fourth direction switching valve **212** are connected to each other, the coolant may flow, and the right side of the fourth direction switching valve **212** may be disconnected. In addition, the upper side and the left side of the fifth direction switching valve **215** may be connected to each other, and the right side of the fifth direction switching valve **215** may be blocked.

Therefore, the coolant in the heating line **230** sequentially passes through the second pump **232**, the coolant heater **233**, the heater core **234**, the first direction switching valve **231**, and the condenser **120**, flows into the second pump **232** again, and circulates. This cycle is repeated. In this case, the coolant passes through the heater core **234** while exchanging heat with the air blown by the air blower of the air conditioning device **190**, such that the air is heat. The heated air is supplied to the vehicle interior and used to heat the vehicle interior.

Further, the coolant in the refrigerating line **210** separated from the heating line **230** flows from the fourth pump **252** to the first direction switching valve **231**, the electrical component **253**, the third direction switching valve **261**, the fourth coolant joint **272**, the chiller **160**, the fifth direction switching valve **215**, the radiator **211**, and the fourth direction switching valve **212**, flows into the fourth pump **252**, and then circulates. This cycle is repeated.

In this case, the upper side and the right side of the third direction switching valve **261** may be connected to each other so that the coolant flows to the chiller **160**. The coolant flows to the chiller **160** via the fourth coolant joint **272**.

Therefore, the electrical component **253** and the chiller **160** may be connected in series, and a coolant flow path capable of absorbing both outside air and waste heat may be formed. In this structure, since there is no branch part in the refrigerating line **210** circulating through the radiator **211**, the electrical component, and the chiller **160**, the coolant may flow at an almost constant flow rate, which may improve the heating performance.

In this case, the coolant may not flow to the battery **213** through the fourth direction switching valve **212** and the fifth direction switching valve **215**.

FIG. **24** is a view illustrating an operating state of the system in the cooling mode.

Referring to FIG. **24**, in the refrigerant circulation line **100**, the compressor **110** operates, and the high-temperature, high-pressure refrigerant is discharged from the compressor **110**. Further, the refrigerant discharged from the compressor **110** is cooled while exchanging heat with the coolant in the condenser **120**. Thereafter, the refrigerant cooled and condensed in the water-cooled condenser **120** is divided in the refrigerant branch part **101**. A part of the refrigerant passes through the refrigerant heat exchanger **180** and then be throttled and expanded while passing through the first expansion valve **130**. Thereafter, the expanded refrigerant exchanges heat with the coolant in the cooling line **220** while passing through the evaporator **140**, and the coolant in the cooling line **220** is cooled by the refrigerant.

Further, the refrigerant evaporated in the evaporator **140** passes through the refrigerant heat exchanger **180**, exchanges heat with the refrigerant before the refrigerant is introduced into the first expansion valve **130**, and then flows into the compressor **110** again via the accumulator **170**.

In addition, the remaining part of refrigerant divided in the refrigerant branch part **101** is throttled and expanded while passing through the second expansion valve **160**. Thereafter, the expanded refrigerant is evaporated by

exchanging heat with the coolant while passing through the chiller **160**, such that the coolant may be cooled. Further, the refrigerant evaporated in the chiller **160** flows into the compressor **110** again via the accumulator **170**.

As described above, the refrigerant having passed through the evaporator **140** and the refrigerant having passed through the chiller **160** merge with each other in the accumulator **170** and flow into the compressor **110**. The refrigerant circulates as the above-mentioned process is repeated.

In addition, the coolant is circulated in the cooling line **220** by the operation of the third pump **221**. Further, the coolant passes through the cabin cooler **222** while exchanging heat with the air blown by the air blower of the air conditioning device **190**, such that the air is cooled. The cooled air is supplied to the vehicle interior and used to cool the vehicle interior.

Meanwhile, the coolant in the coolant circulation line **200** is circulated by the operations of the second pump **232**, the fourth pump **252**, and the first pump **214**. Further, the battery **213**, the electrical component **253**, and the refrigerant passing through the condenser **120** may be cooled by the coolant. The heated coolant may be cooled while exchanging heat with the outside air by the operation of the cooling fan **211a** of the radiator **211** for the electrical component.

In this case, the first direction switching valve **231** and the fourth direction switching valve **212** may adjust the directions thereof so that the heating line **230** and the refrigerating line **210** are connected. More specifically, the coolant may flow as the upper side and the left side of the first direction switching valve **231** are connected to each other, and the coolant may flow as the lower side and the right side of the first direction switching valve **231** are connected to each other. Further, when the left side and the lower side of the fourth direction switching valve **212** are connected to each other, the coolant may flow, and the right side of the fourth direction switching valve **212** may be disconnected. In addition, the upper side and the right side of the fifth direction switching valve **215** may be connected to each other, and the left side of the fifth direction switching valve **215** may be blocked.

Therefore, the coolant flows from the radiator **211** for the electrical component sequentially to the fourth direction switching valve **212**, the fourth pump **252**, the first direction switching valve **231**, the condenser **120**, the second pump **232**, the coolant heater **233**, the heater core **234**, the first direction switching valve **231**, the electrical component **253**, and the sixth coolant joint **262**, flows into the radiator **211** for the electrical component again, and circulates. This cycle is repeated.

In this case, the right line of the third direction switching valve **261** may be closed, thereby preventing the coolant from flowing to the fourth connection line **280**.

The fourth direction switching valve **212** may prevent the coolant from flowing from the fourth direction switching valve **212** to the fifth coolant joint **271**, and the fifth direction switching valve **215** may prevent the coolant from flowing from the fifth direction switching valve **215** to the sixth coolant joint **262**.

In addition, the coolant may flow from the chiller **160** sequentially to the fifth coolant joint **271**, the first pump **214**, the battery **213**, and the fifth direction switching valve **215**, flows into the chiller **160** again, and circulates. This cycle is repeated. That is, the battery **213** and the chiller **160** may define a separate closed loop, in which the coolant circulates, in the refrigerating line **210** by the fourth direction switching valve **212** and the fifth direction switching valve **215**, such that the battery **213** may be separately cooled.

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FIG. 25 is a view illustrating an operating state of the system in a general heating mode.

Referring to FIG. 25, in the refrigerant circulation line 100, the compressor 110 operates, and the high-temperature, high-pressure refrigerant is discharged from the compressor 110. Further, the refrigerant discharged from the compressor 110 is cooled while exchanging heat with the coolant in the condenser 120.

The first expansion valve 130 is closed, such that the refrigerant does not pass through the first refrigerant line 100a in which the refrigerant heat exchanger 180 and the evaporator 140 are disposed. The refrigerant passes through the refrigerant branch part 101 and the third expansion valve and flows into the chiller 160. In the chiller 160, the refrigerant and the coolant exchange heat with each other, such that the refrigerant may be heated.

The refrigerant having passed through the chiller 160 flows into the compressor 110 again via the accumulator 170.

In addition, in the cooling line 220, the third pump 221 does not operate, such that the coolant may not flow in the cooling line 220.

Meanwhile, the coolant in the coolant circulation line 200 is circulated by the operations of the second pump 232, the fourth pump 252, and the first pump 214. Further, the coolant may be heated while passing through the condenser 120, heated by the coolant heater 233, and heated by waste heat of the electrical component 253 and the battery 213. The coolant may be cooled while passing through the chiller 160.

In this case, the first direction switching valve 231 and the fourth direction switching valve 212 may adjust the directions thereof so that the heating line 230 and the refrigerating line 210 are separated. More specifically, the coolant may flow as the upper side and the right side of the first direction switching valve 231 are connected to each other, and the coolant may flow as the lower side and the left side of the first direction switching valve 231 are connected to each other. Further, when the right side and the lower side of the fourth direction switching valve 212 are connected to each other, the coolant may flow, and the left side of the fourth direction switching valve 212 may be disconnected.

Therefore, the coolant in the heating line 230 sequentially passes through the second pump 232, the coolant heater 233, the heater core 234, the first direction switching valve 231, and the condenser 120, flows into the second pump 232 again, and circulates. This cycle is repeated.

Further, the coolant in the refrigerating line 210 separated from the heating line 230 flows from the fourth pump 252 to the first direction switching valve 231, the electrical component 253, the sixth coolant joint 262, and the fifth direction switching valve 215. All the upper side, the left side, and the right side of the fifth direction switching valve 215 may be connected.

In this case, the right line of the third direction switching valve 261 may be closed, thereby preventing the coolant from flowing to the fourth connection line 280.

The refrigerant circulating through the first pump 214 flows to the fifth direction switching valve 215 via the battery 213, merges with the refrigerant having passed through the electrical component 253, and then flows to the chiller 160. The coolant cooled in the chiller 160 is divided in the fifth coolant joint 271 and then flows.

The embodiment has been described in which the coolant is cooled by the water-cooled evaporator 140 as a cooling source and then the vehicle interior is cooled by the cabin cooler 222. However, as another embodiment, it is possible

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to apply a general air conditioner structure in which the evaporator 140 directly exchanges heat with air, and the cooled air flows into the vehicle interior. As a result, the present invention may be applied to the structure of the embodiment in which the cabin cooler 222 and the coolant line through which the coolant flows between the cabin cooler 222 and the water-cooled evaporator 140 are eliminated.

The embodiment of the present invention has been specifically described above with reference to the accompanying drawings.

The above description is simply given for illustratively describing the technical spirit of the present invention, and those skilled in the art to which the present invention pertains will appreciate that various modifications, changes, and substitutions are possible without departing from the essential characteristic of the present invention. Accordingly, the embodiments disclosed in the present invention and the accompanying drawings are intended not to limit but to describe the technical spirit of the present invention, and the scope of the technical spirit of the present invention is not limited by the embodiments and the accompanying drawings. The protective scope of the present invention should be construed based on the following claims, and all the technical spirit in the equivalent scope thereto should be construed as falling within the scope of the present invention.

The invention claimed is:

1. A vehicle heat pump system comprising:

- a compressor configured to compress and circulate a refrigerant;
- a first heat exchanger configured to condense the compressed refrigerant;
- a first expansion valve configured to expand the condensed refrigerant;
- a second heat exchanger configured to evaporate the refrigerant expanded by the first expansion valve by allowing the refrigerant to exchange heat with a coolant;

a cabin cooler configured to cool a vehicle interior by allowing the coolant having passed through the second heat exchanger to exchange heat with air;

a heat-dissipation heat exchanger configured to allow the coolant for cooling an electrical component to exchange heat with outside air,

wherein the coolant having passed through the heat-dissipation heat exchanger flows into the second heat exchanger in a heating mode,

wherein the refrigerant passes through the first heat exchanger, and the vehicle heat pump system further comprises a first refrigerant line branching from a refrigerant branch to one side and passing through the first expansion valve and the second heat exchanger, and a second refrigerant line branching from the refrigerant branch to an other side and passing through a second expansion valve and a chiller,

a heating line configured to heat the vehicle interior by circulating the coolant that exchanges heat with the refrigerant through the first heat exchanger; and

a refrigerating line configured to cool a heat generating component and the electrical component by circulating the coolant that exchanges heat with air or the refrigerant and wherein a first coolant joint and a second coolant joint are disposed in the refrigerating line, such that a first line of the refrigerating line is configured such that the coolant circulates through the heat-dissipation heat exchanger and the second heat exchanger

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and wherein the first line of the refrigerating line is connected to a cooling line configured to cool the vehicle interior by circulating the coolant that exchanges heat with the refrigerant through the second heat exchanger, and the cabin cooler is disposed in the cooling line and a third coolant joint disposed at a side adjacent to an inlet of the second heat exchanger and, the vehicle heat pump system further comprising: a first connection line branching from a first side of the refrigerating line and connecting with the heating line; and a second connection line branching from a second side of the refrigerating line and connecting with the heating line, wherein the first connection line, the second connection line, and the heating line are connected to a first direction switching valve, and wherein the refrigerating line and the heating line are connected or disconnected from each other by the first direction switching valve, wherein the electrical component is disposed on the second connection line, and a third connection line, which branches from a fourth coolant joint disposed on the refrigerating line, is disposed to pass through the chiller, and coolant can flow to the chiller after passing through the electrical component in the heating mode.

2. The vehicle heat pump system of claim 1, wherein the third coolant joint configured to connect the cooling line is disposed in the first line of the refrigerating line, and a second direction switching valve configured to determine whether to circulate the coolant to the cooling line is disposed in the first line of the refrigerating line.

3. The vehicle heat pump system of claim 2, wherein the cooling line comprises a pump connected between the second heat exchanger and the cabin cooler, the pump being disposed between the third coolant joint and the second heat exchanger.

4. The vehicle heat pump system of claim 2, wherein in the heating mode, the second direction switching valve blocks the flow of the cooling line.

5. The vehicle heat pump system of claim 2, characterized in that in the cooling mode, the second direction switching valve controls the direction of coolant so that coolant circulates the cooling line in a closed loop.

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6. The vehicle heat pump system of claim 1, comprising: the second expansion valve configured to expand the condensed refrigerant;

the chiller configured to allow the refrigerant expanded by the second expansion valve to exchange heat with the coolant; and

the electrical component configured to be cooled by the coolant,

wherein in the heating mode, the coolant having passed through the heat-dissipation heat exchanger exchanges heat with the electrical component and then flows into the chiller.

7. The vehicle heat pump system of claim 6, wherein another direction switching valve is disposed in the second connection line and controls a flow direction of the coolant having passed through the electrical component, the another direction switching valve is connected to a fourth coolant joint of the third connection line through a fourth connection line, and the fourth coolant joint is disposed at a side adjacent to an inlet of the chiller.

8. The vehicle heat pump system of claim 1, wherein the second connection line is provided with another direction switching valve for controlling the direction of movement of coolant through the electrical component, the another direction switching valve being connected via the fourth connection line to a fourth coolant joint of the third connection line, the fourth coolant joint being disposed on the inlet side of the chiller.

9. The vehicle heat pump system of claim 1, wherein the first refrigerant line includes a refrigerant heat exchanger disposed between the refrigerant branch and the first expansion valve, wherein refrigerant entering the first expansion valve and refrigerant passing through the second heat exchanger are heat exchanged.

10. The vehicle heat pump system of claim 1, wherein the heating line comprises: a heater core for heating a passenger compartment using air heated by thermal exchange of air entering the passenger compartment with refrigerant through the first heat exchanger; and a coolant heater disposed in front of the heater core in the direction of flow of the coolant to heat the coolant.

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